Citation Paraphrase Verification Log

Thesis: *Abstraction Mechanisms for Virtual Avatars*
Nidanur Günay • University of Konstanz • 2026

Entries ordered by first appearance in the thesis (Ch. 1 → Ch. 4).
Check each ☐ once you have opened the original PDF, located the passage, and confirmed the paraphrase is accurate.

ORANGE NOTE = original text not yet verified from PDF. **RED TEXT** = confirmed original sentence.

MANDATORY WORKFLOW RULE — Citation Entry Protocol

Every time a new citation is added anywhere in the thesis, an entry **must** be added to this document before moving on. The entry must include:

- The **exact sentence** from the thesis where the citation appears (copied verbatim).
- The **section** where it appears (e.g. §4.3, §2.1).
- The **original text** from the source PDF (or an ORANGE "NEEDS VERIFICATION" note if the PDF has not yet been read).
- A **notes** field stating whether the citation is correctly matched to the claim.

The ☐ **Validated** checkbox at the bottom of each entry is confirmed by Nidanur after opening the original PDF, locating the passage, and confirming the paraphrase is accurate.

Do not skip this step. A citation added without a corresponding entry here is unverified and must be treated as a potential error until logged.

Chapter 1 — Introduction

#1 weidner2023 — Weidner et al. (2023) *§1.1 Motivation — new opening paragraph, avatars/agents in immersive environments*

Cited sentence (claim in thesis):	"As virtual assistants, game characters, and avatars in immersive environments increasingly involve users in sustained interactions with virtual humans, users are no longer limited to passive viewing."
Paraphrase used:	Grounds the claim that engagement with avatars and agents in AR/VR is now widespread and direct, not just passive, motivating why rendering-style choice matters more. Sole usage of this source in Chapter 1 as of 2026-08-01 — an earlier draft also cited it a second time in the "Communication through avatars" paragraph for the 72-study systematic-review finding, but that sentence was removed at Nidanur's request as too Related-Work-like for the Introduction; the same finding remains properly cited in `02_related_work.tex` §2.2 (see entry further below).
Original source text:	"AR and VR are pushing from the labs towards consumers, especially with social applications... many popular AR/VR applications use avatars or agents. While an avatar is controlled by a human, an agent is controlled solely by a computer system. (Abstract and §1 Introduction, direct quotes.)"
Notes:	Source: `Articles/_2022_10_07__IEEE_VR__A_Survey_on_Avatar_and_Agents.pdf`. CONFIRMED — read directly, 2026-08-01. NOTE: this replaces a previously fabricated bib entry for `weidner2023` (a nonexistent CHI paper titled "Watching Yourself in a Video Conference," whose DOI actually belonged to an unrelated paper by different authors). The bib entry, and every sentence citing `weidner2023` in `01_introduction.tex` and `02_related_work.tex`, was corrected 2026-08-01 to the real paper: Weidner, Boettcher, Arevalo Arboleda, Diao, Sinani, Kunert, Gerhardt, Broll & Raake, "A Systematic Review on the Visualization of Avatars and Agents in AR & VR Displayed Using Head-Mounted Displays," IEEE TVCG 2023, DOI 10.1109/TVCG.2023.3247072.

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#2 mcdonnell2012 — McDonnell et al. (2012) *§1.1 Motivation — new opening paragraph, paraphrase of the paper's own opening sentence*

Cited sentence (claim in thesis): "These effects raise several design concerns with consequences across film, games, and other domains where virtual humans appear, and one of the most difficult is how realistically or abstractly such a character should be rendered. As a result, this tension between realism and abstraction is the starting point for the work presented in this thesis."

Paraphrase used: Paraphrase (restructured, not a synonym-swap) of the paper's own first sentence, used to establish that render-style choice is a longstanding, recognised difficulty rather than a novel claim invented for this thesis. **UPDATED 2026-08-06:** the two thesis sentences paraphrasing this single source sentence were merged into one, so the citation appears once rather than twice for the same source.

Original source text: "Choosing an appropriate rendering style for virtual humans in movies, games, and other domains can be a challenging task. Using near-human characters can be considered risky, as it is not only costly but can produce a negative audience reaction [Geller 2008; Levi 2004]. (p. 91:1, Introduction, first two sentences.)"

Notes: Source: `Articles/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`. CONFIRMED — Introduction (p. 91:1) read directly, 2026-08-01. **ATTRIBUTION FLAG (resolved):** McDonnell et al.'s second sentence ("near-human characters. . . risky. . . negative audience reaction") is itself credited by them to Geller (2008) and Levi (2004), not an original McDonnell finding — the `[Geller 2008; Levi 2004]` marker was missing from this quote in an earlier pass of this log and has now been restored so the excerpt is not misread as McDonnell's own uncited claim. The actual thesis sentence at `01_introduction.tex` line 8 paraphrases only the first sentence (the general "challenging task" framing) and does not rely on the Geller/Levi-sourced claim, so `mcdonnell2012` alone is the correct and sufficient citation for what the thesis states. Geller (2008) and Levi (2004) do not need their own `references.bib` entries unless a future revision extends the thesis sentence to assert the specific "risky/costly/negative reaction" claim directly. This resolves the outstanding verification flagged in the entry below (same PDF, same file) for at least the opening framing; the render-style/eeriness finding used elsewhere in §1.1 and §2.1 still needs the later results sections read.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#3 natarajan2020 — Natarajan & Gombolay (2020) *§1.1*

Motivation — direct empirical source added 2026-08-02. Sole citation on this sentence — originally co-cited with `alipour2025`*, which was found to be a secondary source citing this same study rather than an independent finding, so* `alipour2025` *was dropped from this sentence and later removed from this log entirely (2026-08-06 cleanup) since it is not used anywhere in Chapter 1.*

Cited sentence (claim in thesis):	"This design choice can be represented along a fidelity axis. Higher fidelity corresponds to greater human likeness, where human likeness refers to the degree to which an avatar visually resembles a real human. Furthermore, an avatar's design and behaviour have been shown to increase users' confidence in its guidance and their willingness to follow it. However, this increase in confidence does not scale linearly with likeness across the fidelity axis."
Paraphrase used:	Identified after Nidanur flagged (2026-08-02) that `alipour2025`'s "Original source text" quote is itself Alipour et al. citing further sources for this exact claim, not their own finding. Natarajan & Gombolay is the concrete empirical study Alipour et al. elaborate on in their own body text (pp. 9–10) as the source of the compliance/trust claim, so it is cited directly rather than only via Alipour et al.'s synthesis.
Original source text:	"Abstract: "Results from the 4x4x2x2 study show that behavior and anthropomorphism of the agent are the most significant factors in predicting the trust and compliance with the robot." §3.4 H1: "The perceived anthropomorphism of an agent will have a positive correlation with the user's trust... we hypothesize that increased perception of anthropomorphism of an agent by users will lead to an increase in compliance with the agent's recommendations and overall trust in the agent." §4.1 Results: "Results from repeated measures ANOVA (rANOVA) show high statistical significance for perceived anthropomorphism ($p<0.001$, $F(1, 720)=17.6276$). Further, we obtained a positive coefficient (0.29335) for anthropomorphism... indicating that trust and anthropomorphism are positively correlated." §4.2 H2 (behavior): "behavior of the agent plays a significant role in affecting user's trust" ($p<0.001$, $F(4,670)=73.5384$)."
Notes:	Source: `Articles/Natarajan-Effects of Anthropomorphism and Accountability on Trust inHuman Robot Interaction.pdf` (physical PDF located and added by Nidanur 2026-08-02). CONFIRMED — read directly, 2026-08-02. Bibliographic details verified from the paper's own header/footer: HRI '20, March 23–26 2020, Cambridge UK, p. 33 (start), ACM ISBN 978-1-4503-6746-2/20/03, DOI 10.1145/3319502.3374839 — all match `references.bib`. H1 (anthropomorphism → trust, explicitly including "compliance with the agent's recommendations" in the hypothesis text) and H2 (behavior → trust) are both statistically confirmed with reported test statistics. **MISMATCH IDENTIFIED AND RESOLVED 2026-08-06:** the thesis sentence previously said "realistic appearance," but Natarajan & Gombolay's H1 variable is anthropomorphism (perceived human-like qualities of a physical robot), not photorealistic rendering fidelity. Anthropomorphism and human likeness/fidelity, as this thesis's own §1.1 paragraph defines the latter ("the degree to which an avatar visually resembles a real human"), are not the same construct: a stylised, non-photorealistic character can still be highly anthropomorphic. The thesis sentence was reworded to the construct-neutral "an avatar's design and behaviour," which is accurate to what H1/H2 actually measured without implying a specific link to photorealism. The paragraph's subsequent "However" sentence was also reworded to pick up this claim explicitly ("this increase in confidence does not scale linearly with likeness") so the contrast into the uncanny valley material still reads coherently.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#4 mori1970 — Mori (1970)	*§1.1 Motivation — first use*
Cited sentence (claim in thesis):	"Mori termed this effect the uncanny valley: comfort rises with human likeness only up to a point, then drops sharply for entities that appear almost but not quite human, before recovering near perfect likeness."
Paraphrase used:	Paraphrase of Mori's original uncanny valley hypothesis. **REVISED 2026-08-01:** the thesis sentence previously said Mori's claim was about "a human representation approaching but falling just short of true photorealistic fidelity," but Mori's original 1970 paper and graph are about robots, prosthetic hands, puppets, and corpses, not virtual or photorealistic representations at all (flagged by Nidanur). Rewritten to state Mori's claim generically (human likeness vs. comfort), with the avatar-specific application moved to the following sentence, which is the thesis's own extension of the concept rather than something attributed to Mori.
Original source text:	"p. 1 ("Valley of familiarity"): "I have noticed that, as robots appear more humanlike, our sense of their familiarity increases until we come to a valley. I call this relation the 'uncanny valley.'" Figure 1 (the familiarity-vs-human-likeness graph, x-axis 0–100% human likeness): the curve rises from industrial robot through humanoid robot/stuffed animal, drops sharply through the trough (labelled "corpse" and "prosthetic hand," positioned close to the 100% end of the axis), then rises again to "healthy person" at 100% human likeness, the highest point on the graph (the "second peak"). p. 2 ("The significance of the uncanny"): "In Figure 1, a healthy person is at the top of the second peak.""
Notes:	Source: `Articles/uncanny1970.pdf`, pp. 1–2. CONFIRMED — read directly in full, 2026-08-06. **TRANSLATION CORRECTION:** the quote previously logged here ("I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases...") does not match the actual local PDF file, which is the MacDorman & Minato (2005) translation, not the "IEEE Spectrum" MacDorman & Kageki translation the old note claimed. The quote above is verified verbatim from the file that is actually saved locally. **Paraphrase accuracy, checked clause by clause:** (1) "comfort rises with human likeness only up to a point" — supported, though Mori's own term is "familiarity," not "comfort"; the thesis substitutes a near-synonym, which is a minor but real terminology shift worth being aware of. (2) "drops sharply for entities that appear almost but not quite human" — supported directly by Figure 1, where the trough (corpse, prosthetic hand) sits close to the 100% end of the human-likeness axis, i.e. near-human rather than mid-range. (3) "before recovering near perfect likeness" — previously unverified in this log (no quote had been logged for it); now confirmed both graphically (the curve's second, higher peak sits at exactly 100% human likeness) and in prose ("a healthy person is at the top of the second peak"). All three clauses of the thesis paraphrase are accurate and supported.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#5 mcdonnell2012 — McDonnell et al. (2012)	*§1.1 Motivation*
Cited sentence (claim in thesis):	"Consequently, an avatar that approaches full realism may reduce users' confidence rather than enhance it. This holds for CG characters generally."
Paraphrase used:	Citation for empirical corroboration of uncanny valley in CG characters. Author name removed 2026-08-01 per Nidanur's request that this read as natural text rather than Related-Work-style attribution. **NOTE:** this is the first half of a two-citation compound sentence — it continues ", and it extends specifically to avatar stylisation and trust [**?**]." The second half is covered by its own separate entry immediately below, since `canales2024` supports only that clause, not this one.
Original source text:	"§7 Discussion (p. 91:10, the clearest single framing sentence): "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale." §5.1.3 Results, Trustworthy ratings (pp. 91:6–7): "HumanIll was considered significantly less trustworthy than all other styles ($p<0.004$ in all cases). HumanBasic was the next least trustworthy ($p<0.004$ in all cases). ToonBare and HumanSSS were next ($p<0.002$ in all cases). Finally, the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles ($p<0.004$ in all cases)." Same pattern for eeriness (p. 91:6): "HumanIll was rated as the most eerie, and significantly more so than all other styles... HumanBasic, ToonBare, and HumanSSS were rated as the next most eerie.""
Notes:	Source: `Articles/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`. CONFIRMED — read directly, 2026-08-02. **Where to look:** the single cleanest supporting sentence is in §7 Discussion, PDF page 10 (printed p. 91:10): "These results are very much in line with the Uncanny Valley theory." The detailed statistical support behind that summary sentence is in §5.1.3 Results, PDF pages 6–7 (printed pp. 91:6–7): the mid-realism styles (HumanIll, HumanBasic, HumanSSS, ToonBare) were rated significantly less trustworthy and more eerie than *both* the fully abstract Toon styles *and* the most realistic Human styles (HumanHQ1/HumanHQ2), which were rated equally and most trustworthy — a genuine valley-shaped (non-monotonic) pattern in trust as a function of realism. **Nuance flagged for honesty:** the Discussion quote itself immediately qualifies this — their most realistic style was rated *as appealing* as their most abstract style (not less), so the appeal/pleasantness dip they found was centred on mid-scale, unfamiliar styles rather than specifically the "almost but not quite realistic" tier Mori's original curve predicts. The *trust*-specific results (not appeal) are the closer match to the thesis's specific claim, since HumanBasic/HumanSSS sit below the top realism tier and were significantly less trusted than it. Caveat: HumanIll was deliberately designed to look ill/diseased (desaturated skin, yellow hue, waxy shading — §4 style descriptions, p. 91:4), a partial confound on its own scores, but HumanBasic/HumanSSS show the same lower-trust pattern without that confound.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#6 canales2024 — Canales et al. (2024) *§1.1 Motivation — **RESOLVED***
***2026-08-08:** sentence reworded by Nidanur to match the actual finding, no longer claims a trust-reduction extension.*

Cited sentence (claim in thesis):	"This holds for CG characters generally. Avatars fall within this category, and their stylisation can affect users' behavioural choice. Together, these results suggest that rendering style influences not only users' subjective confidence but also the decisions they make."
Paraphrase used:	Citation establishing avatar stylisation and behavioural choice as the specific research context. Author name removed 2026-08-01, same rationale as the mcdonnell2012 entry above.
Original source text:	"Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for the second opinion.""
Notes:	**RESOLVED (was MISMATCH, flagged 2026-08-02):** the previous wording claimed the uncanny-valley trust-reduction pattern "extends specifically to avatar stylisation and trust," but Canales et al. found self-reported trust (Ability/Benevolence/Integrity composite, Mayer et al. model) showed *no significant difference* across Real/Mid/Caricature stylisation ($p = .52$), i.e. a null result, alongside a behavioural preference for the Mid stylisation level in an actual choice task. **2026-08-08:** the thesis sentence was reworded (by Nidanur, in her own words, per the citation-check-log-only rule) to claim only that "their stylisation can affect users' behavioural choice," which matches the behavioural-choice finding directly and no longer asserts a trust-reduction extension. This is an accurate, appropriately narrow claim. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule, even though this independent check found the new wording accurate.

Validated — original PDF read; paraphrase confirmed accurate.

#7 rahman2024 — Rahman et al. (2024) *§1.1 Motivation — new replacement citation, 2026-08-02*

Cited sentence (claim in thesis):	"This question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and advertising."
Paraphrase used:	Replaces `weidner2023` in this sentence after that citation was found not to support the claim (see entry above). Rahman et al. is a systematic literature review of avatar-based marketing (57 articles, 2005–2023) that gives concrete, named, real-world examples of avatars in commercial deployment, directly supporting both halves of the sentence (gaming and advertising) in a single citation.
Original source text:	"p. 2 of 46: "Companies like Nike have introduced Nikeland in the Metaverse through platforms like Roblox. Similarly, IKEA has introduced a customer-focused avatar named Anna, who can address inquiries regarding IKEA's products, services, and operating hours." Same page: "...the growing importance of avatars in online services (e.g., banking, gaming, education, shopping, m-health, etc.)... the usage of avatars is not limited to a particular sector, like banking or hotel; it is seen as a huge prospect in the virtual gaming, smartphone, and fashion industries." Same page (advertising half): "...the famous fashion brand Calvin Klein has deployed Lil Miquela, a virtual influencer (another form of avatar); interestingly, Lil Miquela has more than 2.4 million followers on Instagram.""
Notes:	Source: `Articles/Navigating_Avatar-Based_Marketing_From_Value_Aspec.pdf` (physical PDF located and added by Nidanur 2026-08-02), pp. 1–2 of 46, Abstract and Introduction. CONFIRMED — read directly, 2026-08-02. Bibliographic details (Rahman, Sabbir, Zhang, Taufique, Rahman; *International Journal of Consumer Studies*, 48(6), e13096; DOI 10.1111/ijcs.13096) verified from the paper's own title page, matching `references.bib` exactly. Unlike the replaced `weidner2023` citation, these are real, named commercial deployments (Nike/Roblox for gaming, Calvin Klein's Lil Miquela virtual influencer for advertising/marketing), not academic lab studies, so this is a materially stronger match for the claim.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#8 canales2024 — Canales et al. (2024) *§1.2 Problem Statement — new usage 2026-08-06, literature-gap paragraph.* ***RESOLVED 2026-08-08:*** *over-generalisation fixed.*

Cited sentence (claim in thesis):	"A recent avatar stylisation study focused on perceptual outcomes rather than the practical challenges of production integration."
Paraphrase used:	Cited as an example of avatar-stylisation research that measures perceptual and behavioural outcomes (self-reported trust ratings, a forced-choice doctor-selection task) rather than production engineering integration challenges, supporting the claim that this branch of the literature has not addressed the integration problem this thesis investigates. **2026-08-08:** "Existing work on avatar stylisation" was narrowed to "A recent avatar stylisation study," matching the fact that only one paper (a single perceptual study, not a survey) is cited.
Original source text:	"Abstract: "Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion." Introduction: "Results show very little impact of stylization. Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy. Participants were more likely to select the Mid level of stylization when picking a doctor for a second opinion." Methods, explicit self-description: "In order to ensure behavioral consistency across participants and to limit the technical complexity, we used prerecorded avatar animation rather than live tracking.""
Notes:	Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. CONFIRMED — full-text searched and relevant sections read directly, 2026-08-06 (zero hits anywhere in the 15-page paper for "production," "implementation," "render pipeline," "shader," or "integration"). The paper's own Methods section explicitly states they simplified away technical/production complexity (prerecorded animation, not live tracking) specifically to keep focus on the perceptual question, directly supporting the "perceptual outcomes rather than production integration" characterisation. **RESOLVED 2026-08-08:** the over-generalisation flagged 2026-08-06 ("Existing work" implying a literature-wide pattern from one paper) is fixed by narrowing the subject to "A recent avatar stylisation study." **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

Validated — original PDF read; paraphrase confirmed accurate.

#9 isenberg2003 — Isenberg et al. (2003) *§1.2 Problem Statement — sentence revised 2026-08-04, paragraph restructured 2026-08-06*

Cited sentence (claim in thesis): "Classical NPR techniques share this same gap, since they are typically demonstrated in rendering environments where developers have direct control over rendering operations and asset creation. It remains unclear how these techniques transfer to avatar platforms with substantially different integration constraints."

Paraphrase used: **REVISED 2026-08-04:** the previous wording ("the literature assumes full pipeline access and custom assets under complete authorial control") was flagged by Nidanur as an unverifiable sweeping claim about the whole NPR literature that Isenberg et al. never explicitly state. The sentence now describes what this specific paper demonstrates rather than what it assumes about the field: a developer-facing guide whose surveyed algorithms require inserting extra rendering passes and manipulating the mesh, z-buffer, or normal buffer directly, illustrated throughout with simple custom test geometry rather than production or photogrammetric assets. This is a narrower, directly supportable claim about this paper's own demonstration context, not an inference about the field at large. **UPDATED 2026-08-06:** the paragraph was restructured to add an explicit bridging sentence ("The NPR literature leaves the same gap unaddressed from the opposite direction," immediately preceding this citation, uncited, the thesis's own synthesis connecting this citation to the `canales2024` claim above it) and the closing sentence was simplified back to a generic field-level statement rather than referencing the thesis directly. The claim this citation supports and its verification are unchanged; only the surrounding connective prose moved.

Original source text: "This article provides a guideline for developers who need to choose one of these algorithms for his or her application. [...] Hybrid algorithms are characterized by operations in object space that are followed by rendering the modified polygons in a more or less ordinary way using the z-buffer. This usually requires two or more rendering passes."

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, pp. 28–30 (Introduction, Classification, Hybrid algorithms). CONFIRMED — read directly, 2026-08-04. The paper is explicitly framed as "A Developer's Guide to Silhouette Algorithms for Polygonal Models," discusses direct manipulation of the z-buffer/normal buffer and insertion of extra rendering passes (object-space and hybrid algorithm classes), and Figures 1–2 demonstrate the surveyed algorithms on simple custom test models (a nut, a cross) rather than production or captured assets — supporting both the "rendering operations" and "asset creation" halves of the revised claim.

Validated — original PDF read; paraphrase confirmed accurate.

#10 zell2015 — Zell et al. (2015) *§1.2 Problem Statement — new usage added 2026-08-06, citation previously only used in §4.3 Methodology (see entry further below) for the same underlying claim.* ***RESOLVED 2026-08-08:*** *wording precision flag fixed.*

Cited sentence (claim in thesis):	"This is because stronger abstraction typically reduces the perceived intensity of facial expressions, even where basic recognition is preserved."
Paraphrase used:	Previously an uncited flat claim, flagged during thesis review. The identical claim ("excessive stylisation can reduce visual detail and compromise expressive readability") is already cited with `zell2015` and `wallraven2008` in `04_methodology.tex` §4.3, so the same pair was added here rather than searching for a new source.
Original source text:	"Main Findings list, p. 184:2: "The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization.""
Notes:	Source: `Articles/NPR/To Stylize or not to Stylize?_Zell.pdf`. RE-CONFIRMED — read directly again, 2026-08-06 (independently of the prior session's confirmation for the §4.3 usage), verbatim quote matches exactly. **RESOLVED 2026-08-08:** the wording precision flag from 2026-08-06 ("reduces legibility" vs. the paper's actual measure of perceived intensity) is fixed — the thesis sentence now says "reduces the perceived intensity of facial expressions, even where basic recognition is preserved," matching the source directly. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#11 wallraven2008 — Wallraven et al. (2008) *§1.2*

Problem Statement — new usage added 2026-08-06, paired with `zell2015` above, same rationale. ***RESOLVED 2026-08-08:*** *wording precision flag fixed.*

Cited sentence (claim in thesis):	"[Same sentence as above, paired citation with zell2015:] This is because stronger abstraction typically reduces the perceived intensity of facial expressions, even where basic recognition is preserved."
Paraphrase used:	Paired with `zell2015` for the same claim, matching the existing §4.3 Methodology pairing.
Original source text:	"§4 Summary, p. 23:19: "In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression.""
Notes:	Source: `Articles/NPR/Evaluating the perceptual realism of animated facial expressions. .pdf`. RE-CONFIRMED — read directly again, 2026-08-06, verbatim quote matches exactly. **RESOLVED 2026-08-08:** the wording precision flag from 2026-08-06 is fixed — the thesis sentence now states the claim exactly as this pair supports it: reduced perceived intensity, with basic recognition explicitly preserved. This is worded more precisely than the parallel §4.3 Methodology usage of this pair (see entry further below), which still says "compromise expressive readability" and has not been revised. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

Chapter 2 — Related Work

#12 mori1970 — Mori (1970) *§2.1 Uncanny Valley and Avatar Realism — hypothesis definition, reworded 2026-08-07*

Cited sentence (claim in thesis):	"The uncanny valley hypothesis was first articulated by Mori. He proposed that human comfort towards an artificial figure increases with its human likeness. This rise holds only up to a critical threshold. Beyond it, subtle imperfections cause a sharp drop in comfort. The drop persists until the figure appears almost fully human. Only then is comfort recovered."
Paraphrase used:	Paraphrase of the core UV hypothesis. **UPDATED 2026-08-07:** split into shorter sentences and reworded "affinity" to "comfort" throughout to match the figure's own axis label (*shinwakan*/"comfort level"). The citation now sits at the end of the full multi-sentence description of Mori's curve (rise, threshold, drop, recovery) rather than mid-way through it, consistent with how this same pattern was handled in Chapter 1. The claim itself and its source verification are unchanged.
Original source text:	"I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases until we come to a valley (Figure 1), which I call the uncanny valley."
Notes:	CONFIRMED. Same source as Entry #3. The thesis paraphrase adds "subtle imperfections cause a sharp drop" which is also supported by Mori's text about the prosthetic hand and zombie examples.

Validated — original PDF read; paraphrase confirmed accurate.

#13 mori1970 — Mori (1970) *§2.1 Uncanny Valley and Avatar Realism — reference points paragraph, citation added 2026-08-07*

Cited sentence (claim in thesis):	"Mori plots separate curves for still and moving figures. The reference points Mori places inside the valley, a corpse, a zombie, and a prosthetic hand, share a common characteristic. Each looks human in most respects, but each has one flaw that people notice instantly."
Paraphrase used:	This paragraph previously had no citation anywhere despite describing Mori's own figure and examples in detail. Added at the end of the Mori-specific span (before the sentence turns to the thesis's own avatar application), consistent with how the preceding paragraph's citation is placed.
Original source text:	"Figure 1 legend: "moving ——" / "still ———". Figure 1 labels near the valley trough, in order along the curves: "corpse" (still curve), "zombie" (bottom of moving curve), "prosthetic hand" (near 100% on both curves). Prose on the prosthetic hand: "the appearance is quite human like, but the familiarity is negative... we are surprised by the lack of soft tissue and cold temperature." Prose on death/corpse: "our body becomes cold, our color changes, and movement ceases.""
Notes:	Source: `Articles/uncanny1970.pdf` (Mori 1970, trans. MacDorman & Minato), Figure 1 and surrounding prose. CONFIRMED — read directly, 2026-08-07. The two-curves-for-still/moving claim and the three named reference points (corpse, zombie, prosthetic hand) are both directly confirmed from Figure 1. The "each looks human in most respects, but each has one flaw people notice instantly" sentence is a fair synthesis of the prosthetic-hand and corpse descriptions but is not a direct quote from Mori, no single sentence in the source states this as a generalisation across all three reference points. **CAVEAT NOT YET RESOLVED:** the following sentences in the same paragraph give specific percentage ranges ("roughly sixty to seventy percent," "eighty to ninety-five percent") for where the stuffed animal/humanoid robot and the most problematic region sit. Mori's prose gives no percentages at all, only Figure 1's axis is labelled, with tick marks only at 50% and 100%. These specific ranges cannot be confirmed from the text and would need a precise pixel-position reading of Figure 1 itself to verify before citing to `mori1970`. Flagged, not yet resolved.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#14 macdorman2006 — MacDorman & Ishiguro (2006) *§2.1*

Uncanny Valley — empirically tractable research programme. ***RESOLVED 2026-08-09:*** *unsupported semantic-differential claim removed.*

Cited sentence (claim in thesis):	"MacDorman and Ishiguro advanced the uncanny valley from a qualitative hypothesis into an empirically tractable research programme. They extended the theory to androids, which are robots designed to be indistinguishable from humans in appearance and behaviour. It has been argued that these androids offer a uniquely controllable stimulus for cognitive and social science research. This is because androids can be positioned at any point along the human likeness continuum with experimental control."
Paraphrase used:	Paraphrase of MacDorman & Ishiguro's methodological contribution. **RESOLVED 2026-08-09:** full-text search of the paper (`pdftotext` grep for "semantic differential", "questionnaire", "rating scale") found zero mentions of a semantic differential instrument anywhere in the 42-page paper — the "they proposed semantic differential instruments to quantify eeriness and strangeness" claim was removed from the thesis sentence. That methodological contribution actually belongs to `ho2010` (Ho & MacDorman 2010's validated two-subscale eeriness instrument), which the thesis already correctly cites for exactly this in §2.2. The remaining claims (android definition, controllable-stimulus argument) were already confirmed and are unaffected.
Original source text:	"p. 297 abstract: "The development of robots that closely resemble human beings can contribute to cognitive research. An android provides an experimental apparatus that has the potential to be controlled more precisely than any human actor." p. 298: "An android is defined to be 'an artificial system designed with the ultimate goal of being indistinguishable from humans in its external appearance and behavior' (MacDorman & Ishiguro, 2005)."""
Notes:	Source: `Articles/MacDorman2006AndroidScience.pdf`. CONFIRMED — full 42-page text searched directly (2026-08-09) via `pdftotext`, confirming "semantic differential" and "categorical perception" both return zero hits in the body text. The android definition and controllable-stimulus argument remain confirmed from the abstract and p. 298. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

Validated — original PDF read; paraphrase confirmed accurate.

#15 macdorman2006 — MacDorman & Ishiguro (2006) *§2.1 Uncanny Valley — categorical perception mechanism.* ***RESOLVED 2026-08-09:*** *claim removed, not supported by this paper.*

Cited sentence (claim in thesis): "[Claim removed from thesis text 2026-08-09.]"

Paraphrase used: Paraphrase of a specific mechanistic claim (categorical perception as the cause of uncanny valley discomfort) attributed to MacDorman & Ishiguro.

Original source text: "N/A — claim removed."

Notes: Source: `Articles/MacDorman2006AndroidScience.pdf`. CONFIRMED UNSUPPORTED — full 42-page text searched directly (2026-08-09) via `pdftotext` grep for "categorical"; the only hit is an unrelated 1997 reference in the bibliography ("Symbol grounding: Learning categorical and sensorimotor predictions"), not a claim in this paper's own body text. "Categorical perception" as an uncanny valley mechanism is not discussed anywhere in this source. The thesis sentence attributing this mechanism to MacDorman & Ishiguro (2006) was removed and replaced with a claim grounded in Mori's own curve (`mori1970`, already cited earlier in the same section), which does not require a new source. No replacement citation was fabricated for the categorical-perception claim itself; if a real source for it is identified later, it can be added back with proper attribution. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

Validated — original PDF read; paraphrase confirmed accurate.

#16 mcdonnell2012 — McDonnell et al. (2012) *§2.1 Uncanny Valley — render style study.* ***RESOLVED 2026-08-09:*** *factual error and directional mismatch both corrected.*

Cited sentence (claim in thesis): "Participants were exposed to video stimuli of the same animated character, rendered across eleven distinct styles spanning the range from photorealistic to fully stylised. The results showed a valley-shaped pattern across appeal, reassurance, and trustworthiness alike: styles at the most abstract and the most realistic ends of the range scored significantly higher on all three measures than styles in the mid-realism range."

Paraphrase used: Paraphrase of McDonnell et al.'s key finding on render style, appeal, reassurance, and trustworthiness.

Original source text: "§4: "We created 11 rendering styles in total for use in our experiments." §5.1.3 Realism: "Posthoc analysis showed that six distinct levels of realism were found" (this is the source of the previous "six" error — six is the number of statistically grouped realism levels, not the number of styles tested). Appeal: "four of the most abstract and the two most realistic styles were rated as the most appealing... and significantly more appealing than all other styles... characters that were rated halfway between abstract and realistic... were considered the least appealing." Re-assuring: "the most realistic and most abstract styles were considered the most re-assuring... HumanIll was rated as the most eerie." Trustworthy: "the most realistic Humans, and all other Toons were rated as equally trustworthy, and more trustworthy than all other styles." §7 Discussion: "These results are very much in line with the Uncanny Valley theory. Contrary to the theory, however, was the fact that our most realistic character was often rated as appealing or pleasant as our cartoon characters. The drop in appeal for our stimuli came from characters that were rated as unfamiliar or those that were in the middle of our 'abstract to realistic' scale.""

Notes: Source: `Articles/ McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012). Render me Real? Investigating.pdf`. CONFIRMED — full 11-page paper read directly, 2026-08-09. **TWO ERRORS FOUND AND FIXED:** (1) "six distinct styles" was factually wrong — the paper used 11 render styles (10 in the stills block); six is the number of statistically distinct realism levels a post-hoc analysis grouped those styles into, an unrelated number. (2) The previous sentence claimed photorealistic rendering scored significantly worse than stylised conditions on eeriness and appeal, and that NPR styles "consistently" did better. The paper's actual finding is a valley/U-shaped pattern: both the most abstract AND the most realistic styles scored significantly better on appeal, reassurance, and trustworthiness than the mid-realism styles, which include both "Human"-family styles (HumanIll, HumanBasic, HumanSSS) AND one Toon/NPR style (ToonBare) — so it is not accurate that NPR styles uniformly outperformed realistic ones. The corrected sentence reflects the true U-shaped pattern across all three relevant measures (appeal, reassurance/eeriness, and trustworthiness), which is the same pattern the log already correctly identified for a different usage of this citation in Chapter 1. **Not yet starred** — pending Nidanur's explicit confirmation per the re-check-before-starring rule.

Validated — original PDF read; paraphrase confirmed accurate.

#17 seymour2021 — Seymour et al. (2021) *§2.1 Uncanny Valley — field study, fully rewritten and verified 2026-08-08*

Cited sentence (claim in thesis):	"Seymour et al. tested this directly using a professionally produced, real-time photorealistic avatar in a field study. None of the eighteen participants who interacted with it reported an uncanny or aversive response, and the avatar was also rated as having greater affinity, trustworthiness, and preference than a caricature counterpart."
Paraphrase used:	**MISMATCH FOUND AND CORRECTED 2026-08-08:** the previous wording ("the caricature condition did not trigger uncanny valley discomfort, occupying instead a stable neutral region... trust advantages associated with photorealism are driven by factors beyond the mere avoidance of the valley") had the paper's own finding backwards. The paper's headline result, stated in its own title and abstract, is that the *realistic* avatar crossed the uncanny valley, not that it merely won on trust despite residual uncanny risk. The caricature avoiding discomfort is a trivial by-design fact (caricatures are deliberately positioned at low realism, confirmed in the paper's own pilot testing, Table 2), not a notable empirical finding on its own. The corrected sentence states what the paper actually found: zero aversive responses to the realistic avatar, and the realistic avatar winning on affinity, trust, and preference simultaneously.
Original source text:	"Abstract: "Our results suggest that it is now possible to cross the uncanny valley with human-realistic avatars rendered in real time." Title: "Have We Crossed the Uncanny Valley?" Table 5 (Observation Results): "Uncanniness of Digital MIKE: 0 of 18" guests displaying an aversive response. p. 15: "All but three participants preferred the more human-realistic avatar to the caricature avatar." H1–H3 (p. 5): greater affinity, greater trustworthiness, and preference for the human-realistic avatar over the caricature, all supported per the abstract's summary of results."
Notes:	Source: `Articles/Seymour, Yuan, Dennis & Riemer, ''Have We Crossed the Uncanny Valley?'' (2021).pdf` (copied to a working filename to work around special characters in the original filename during reading). CONFIRMED — full paper read directly, 2026-08-08 (pp. 1–15 of 27, covering abstract, theory, hypotheses, methodology, and the qualitative/quantitative results relevant to this claim). Journal of the Association for Information Systems, 22(3), DOI 10.17705/1jais.00491 — matches `references.bib`.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

PDF available. Verify the specific finding about trustworthiness ratings and the caricature not triggering discomfort.

#18 mayer1995 — Mayer et al. (1995) *§2.2 Trust — ABI model definition*

Cited sentence (claim in thesis):	"Mayer et al. defined trustworthiness as comprising three independently evaluable components: ability, benevolence, and integrity."
Paraphrase used:	Paraphrase of the ABI trust model.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Mayer, Davis, & Schoorman (1995), "An integrative model of organizational trust," *Academy of Management Review*, 20(3), 709–734. Key sentence likely from Abstract or p. 717.]"
Notes:	No PDF in `Articles/` folder. This is a foundational paper cited extensively. Obtain the PDF and verify. The ABI model and the definition of trust as "willingness to be vulnerable" are confirmed to be from this paper by the Canales (2024) paper which cites it for the same definition.

Validated — original PDF read; paraphrase confirmed accurate.

#19 mcknight2001 — McKnight & Chervany (2001) *§2.2 Trust — trusting beliefs typology*

Cited sentence (claim in thesis):	"McKnight and Chervany extended this conceptual structure into a typology separating trusting beliefs from trusting intentions, allowing trust measurement to span both attitudinal and behavioural dimensions."
Paraphrase used:	Paraphrase of McKnight & Chervany's typology of trust.
Original source text:	"[NEEDS VERIFICATION — read `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage about trusting beliefs vs. trusting intentions.]"
Notes:	PDF available: `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf`. Verify this is the correct McKnight & Chervany 2001 paper and confirm the distinction between trusting beliefs and intentions.

Validated — original PDF read; paraphrase confirmed accurate.

#20 alipour2025 — Alipour et al. (2025) *§2.2 Trust — systematic review*

Cited sentence (claim in thesis):	"Alipour et al. conducted a PRISMA-compliant systematic review of 53 empirical studies at the intersection of Uncanny Valley effects and trust."
Paraphrase used:	Paraphrase of Alipour et al.'s scope.
Original source text:	"Following PRISMA guidelines, a systematic search identified 53 empirical studies that investigated both UVE-related constructs and trust or trust-related outcomes."
Notes:	Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. CONFIRMED.

Validated — original PDF read; paraphrase confirmed accurate.

#21 alipour2025 — Alipour et al. (2025) *§2.2 Trust — methodological prescription*

Cited sentence (claim in thesis):	"Alipour et al. conclude that well-designed future studies should employ dynamic video stimuli and include at least one behavioural indicator alongside self-report ratings."
Paraphrase used:	Paraphrase of Alipour et al.'s design recommendation.
Original source text:	"The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."
Notes:	Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. PARTIALLY CONFIRMED. The abstract confirms the limitation of static images and subjective measures, which implies the recommendation for dynamic stimuli. Read further pages to locate the exact forward-looking prescription sentence.

Validated — original PDF read; paraphrase confirmed accurate.

#22 alipour2025 — Alipour et al. (2025) *§2.2 Trust — Canales et al. design mapped to the taxonomy*

Cited sentence (claim in thesis):	"This pairing of a self-report scale with a forced-choice behavioural measure spans two of the three measurement categories in the Alipour et al. taxonomy."
Paraphrase used:	Analytical claim: applies the three-category taxonomy (already confirmed in Entry #20 above) to characterise Canales et al.'s measurement design, not a new independent finding from Alipour et al.
Original source text:	"Following PRISMA guidelines... direct trust measures ($n = 24$), trust-related constructs ($n = 11$), and trust-related intentions ($n = 18$)."
Notes:	No new primary-source excerpt needed beyond Entry #20 (same source, same taxonomy). Flag: verify that Canales et al.'s self-report scale and forced-choice task correspond to the "direct trust measures" and "trust-related intentions" categories specifically, rather than the "trust-related constructs" category, before treating this mapping as settled.

Validated — original PDF read; paraphrase confirmed accurate.

#23 nowak2005 — Nowak & Rauh (2005) *§2.2 Trust — avatar anthropomorphism and credibility*

Cited sentence (claim in thesis): "In a study where 255 participants evaluated a series of avatars, perceived anthropomorphism and avatar gender expression both predicted credibility judgments. More anthropomorphic avatars were rated as more credible."

Paraphrase used: Paraphrase of Nowak & Rauh's credibility finding.

Original source text: "p.158: "Participants ($N = 255$) were recruited...". Table 3 (p.170): anthropomorphism correlated with credibility, $r = .46$, $p < .01$. p.172 (Discussion): "A gendered avatar (whether masculine or feminine) was more credible than an androgynous avatar and more likely to be selected." p.174 (Conclusion): "Avatars that were more anthropomorphic were perceived to be more attractive and credible, and people were more likely to choose to be represented by them.""

Notes: Source: `Articles/Nowak Rauh 2005.pdf`, pp.158, 170, 172, 174. CONFIRMED. Note: the credibility regression (RQ8, p.169) found credibility predicted only by animal/object category, not by the anthropomorphism rating directly; the anthropomorphism–credibility link is the correlational finding (r=.46) and the paper's own Conclusion, not a controlled regression effect. The thesis wording matches the paper's own Conclusion phrasing, so this is not a misrepresentation.

✔ **Validated** — original PDF read; paraphrase confirmed accurate.

#24 nowak2003 — Nowak & Biocca (2003) *§2.2 Trust — diminishing returns of anthropomorphism*

Cited sentence (claim in thesis): "Nowak and Biocca extended this framework by demonstrating that anthropomorphism and perceived agency contribute independently to social presence and copresence, and that very high levels of visual anthropomorphism do not consistently produce higher social presence."

Paraphrase used: Paraphrase of the diminishing-returns finding.

Original source text: "Participants interacting with the less-anthropomorphic image reported more copresence and social presence than those interacting with partners represented by either no image at all or by a highly anthropomorphic image of the other, indicating that the more anthropomorphic images set up higher expectations that lead to reduced presence when these expectations were not met."

Notes: Source: `Articles/Nowak2003.pdf`, Abstract. CONFIRMED. The abstract directly states the diminishing-returns pattern.

Validated — original PDF read; paraphrase confirmed accurate.

#25 canales2024 — Canales et al. (2024) *§2.2 Trust — multi-component stylisation study*

Cited sentence (claim in thesis):	"Self-reported trust ratings were broadly equivalent across all three style levels; however, the semi-realistic condition attracted the highest rate of behavioural selection."
Paraphrase used:	Paraphrase of Canales et al.'s key dissociation finding.
Original source text:	"Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion."
Notes:	Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`, Abstract. CONFIRMED. Both halves of the dissociation are in the abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#26 gao2025 — Gao et al. (2025) *§2.2 Trust — stylised vs realistic comparable trust*

Cited sentence (claim in thesis):	"Gao et al. found that stylised agents achieved trust levels statistically comparable to realistic agents on all measured dimensions, while voice characteristics were found to have a negligible effect on trust outcomes."
Paraphrase used:	Paraphrase of Gao et al.'s equivalence finding.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Gao et al. (2025). Verify: (a) trust equivalence across stylisation, (b) voice negligible effect, (c) perceived agent age as moderator.]"
Notes:	No PDF available. The thesis makes three specific claims about this paper (trust equivalence, voice effect, age moderator). All three must be verified.

Validated — original PDF read; paraphrase confirmed accurate.

#27 weidner2023 — Weidner et al. (2023) *§2.2 Trust — rendering style, task dependence*

Cited sentence (claim in thesis): "Weidner et al. synthesised the effect of rendering style at a much larger scale in a systematic review of 72 studies on avatar and agent visualisation in augmented and virtual reality. Full-body, realistically rendered avatars generally produced stronger presence, embodiment, and task performance than partial-body or more abstract alternatives, but this advantage was task dependent. Highly realistic avatars offered little benefit, and were sometimes a hindrance, in game-like and educational scenarios."

Paraphrase used: Establishes at a large, systematically reviewed scale that photorealism's advantages are not unconditional, motivating the thesis's test of whether they hold in a video-based advisory scenario.

Original source text: ""Our findings show that full-body visualizations usually outperform partial body configurations. Also, in most studies, a realistic rendering style yields better results than other rendering styles... we found that the task domain influences avatar selection as for example more task-focused activities (e.g., gaming or education) do not benefit much from highly realistic avatars or are in some cases even hampered by them." (§6 Conclusion, direct quote.)"

Notes: Source: `Articles/_2022_10_07__IEEE_VR__A_Survey_on_Avatar_and_Agents.pdf`. CONFIRMED — Conclusion read directly, 2026-08-01. Corrected replacement for a previously fabricated citation (see §1.1 entries in Chapter 1 above for the full note): the old sentence's claims about video conferencing and identification-mediated fidelity effects are not supported by the real paper and have been removed from `02_related_work.tex`.

Validated — original PDF read; paraphrase confirmed accurate.

#28 bartneck2009 — Bartneck et al. (2009) *§2.2 Trust — Godspeed Questionnaire*

Cited sentence (claim in thesis): "The Godspeed Questionnaire Series, developed by Bartneck et al., provides validated semantic differential scales for five perceptual dimensions: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety."

Paraphrase used: Citation for the Godspeed Questionnaire.

Original source text: "[NEEDS VERIFICATION — read `Articles/trust/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf` and confirm the five subscales and their item descriptions.]"

Notes: PDF available. Confirm the five subscales. Note: the paper's title is "Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots." Confirm this is the Bartneck et al. 2009 Godspeed paper.

Validated — original PDF read; paraphrase confirmed accurate.

#29 macdorman2006 — MacDorman & Ishiguro (2006) *§2.2 Trust — eeriness-specific instruments*

Cited sentence (claim in thesis):	"The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly."
Paraphrase used:	Citation for MacDorman's eeriness operationalisation.
Original source text:	"[NEEDS VERIFICATION — read remaining pages of `Articles/MacDorman2006AndroidScience.pdf` (read pp. 297–300 so far) to locate the specific eeriness instrument or scale description.]"
Notes:	Partially read. The 4 pages read do not yet show the scale. Read the rest of the paper. Note: the thesis may be conflating MacDorman 2006 with a different paper for the "eeriness instrument" — verify that the scale is actually presented here vs. in a later MacDorman paper.

Validated — original PDF read; paraphrase confirmed accurate.

#30 ho2010 — Ho & MacDorman (2010) *§2.2 Trust — eeriness and human-likeness orthogonality*

Cited sentence (claim in thesis):	"Ho and MacDorman... establishing it as a standalone alternative to the Godspeed eeriness items. Their principal finding is that the two dimensions are orthogonal: the reported correlation between eeriness and human-likeness in their validation data is approximately zero ($r \approx .02$)."
Paraphrase used:	Paraphrase of Ho & MacDorman's orthogonality finding.
Original source text:	"Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)."
Notes:	Source: `Articles/trust/Ho2010UncannyValleyIndices.pdf`, Abstract. CONFIRMED. The $r = .02$ figure is in the abstract. Confirm the instrument description (6 items) by reading further pages.

Validated — original PDF read; paraphrase confirmed accurate.

§2.3 Non-Photorealistic Rendering for Virtual Humans

#31 lake2000 — Lake et al. (2000) *§2.3 NPR — toon shading definition*

Cited sentence (claim in thesis):	"Toon shading is one strand of this tradition. It quantises the continuous diffuse lighting term into a small number of discrete bands."
Paraphrase used:	Paraphrase of Lake et al.'s basic toon/cartoon shading description.
Original source text:	"[§3.1 "Cartoon Shading":] a 1D lighting value indexes a small (1–2 texel) gradient texture, producing a small number of discrete, hard-edged tonal bands rather than a continuous Lambertian ramp."
Notes:	RESOLVED 2026-08-11 (previously flagged orange, "resolve before submission"): the prior sentence attributed an inverted-hull outline technique to Lake et al., which a full read of `Articles/NPR/lake et al Stylized rendering.pdf` confirmed is NOT in the paper — Lake et al. describe SED, not backface-normal expansion, and the paper states outline width scales with camera distance, directly contradicting the old "distance-independent" claim. The inverted-hull citation has been replaced with `raskar1999` (see below). `lake2000` is now cited only for the claim it genuinely supports: §3.1's description of toon shading as diffuse-term quantisation into discrete bands. CONFIRMED against the primary PDF.

Validated — original PDF read; paraphrase confirmed accurate.

#32 raskar1999 — Raskar & Cohen (1999) *§2.3 NPR — silhouette rendering via translated back-facing polygons*

Cited sentence (claim in thesis):	"Raskar and Cohen proposed processing the mesh twice from the same viewpoint. Initially, it draws the front-facing polygons in the usual way. The back-facing polygons are then rendered and translated towards the camera."
Paraphrase used:	Paraphrase of the two-render translated-backface silhouette technique.
Original source text:	"[§3.2 "Translated Back-facing Polygons":] front-facing polygons are rendered normally, then back-facing polygons are rendered a second time, translated toward the viewer by a fixed amount t or a depth-scaled amount $k \cdot z$, so the translated back-facing geometry appears in front of the front-facing surface along silhouette boundaries."
Notes:	NEW 2026-08-11. Source: `scratchpad/raskar_cohen_real.pdf`, the author's own copy from `web.media.mit.edu/~raskar/UNC/NPR/sil-i3d99.pdf`, read in full. (A Semantic-Scholar-cached PDF found first was a third-party student summary slide deck, not the primary paper, and was discarded.) Located via Isenberg et al. (2003) Figure 3, credited "Courtesy of Ramesh Raskar and Michael Cohen © 1999 ACM." Not yet starred — pending Nidanur's explicit confirmation.

Validated — original PDF read; paraphrase confirmed accurate.

#33 isenberg2003 — Isenberg et al. (2003) *§2.3 NPR — silhouette edges vs. feature lines*

Cited sentence (claim in thesis):	"Isenberg et al. provided a systematic survey of silhouette detection algorithms. They distinguish silhouette edges from feature lines such as creases, borders, and self-intersections."
Paraphrase used:	Paraphrase of Isenberg's definitional distinction between silhouette edges and feature lines.
Original source text:	"pp. 28–29: silhouette edges mark an object's outer boundary as seen from the current viewpoint. Feature lines (creases, borders, self-intersections) are defined as a separate category of interior edges not detected by silhouette-only algorithms."
Notes:	CORRECTED 2026-08-11: the previous version of this entry quoted "Thus, these geometric methods are not capable of identifying structural boundaries and cannot convey internal surface details" as if verbatim from the source. On rereading the full PDF, no such sentence was found, and the paper's real classification is image-space/object-space/hybrid, not a "geometry-based" category — the earlier excerpt field appears to have been a paraphrase presented as a quote. The thesis sentence has been rewritten to use the paper's own verifiable definitional framework (silhouette edges vs. feature lines) instead. CONFIRMED against `Articles/NPR/Isenberg_2003_ADG.pdf` in full, not a secondary paraphrase. Note: `isenberg2003` is also cited several other places in the thesis (Introduction, Methodology) for unrelated claims — those other citations were not touched by this correction and were not re-verified as part of this update.

Validated — original PDF read; paraphrase confirmed accurate.

#34 gonzalez2018 — Gonzalez & Woods (2018) *§2.3 NPR — Sobel for luminance boundaries*

Cited sentence (claim in thesis): "The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours, and painted surface details."

Paraphrase used: Citation for Sobel operator and gradient mathematics.

Original source text: "[p. 718, §10.2 "Gradient Operators":] "Obtaining the gradient of an image requires computing the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ at every pixel location in the image." [p. 719, Eqs. (10-24) and (10-25):] $g_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$ and $g_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$ [Sobel kernels with ±2 weighting in centre row/column, as shown in Fig. 10.14.] "These kernels are called the *Sobel operators* (Sobel [1970])." [p. 719:] "The fact that the Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable.""

Notes: Source: `Articles/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, pp. 718–719. CONFIRMED. The G_x/G_y matrices with ±2 centre weighting are confirmed in Fig. 10.14 and Eqs. (10-24)–(10-25). NOTE: BT.601 luma coefficients still need separate verification in the colour chapter (Ch. 6). Do not mark the BT.601 claim as confirmed until that is done. ATTRIBUTION CHECK: the quote's "(Sobel [1970])" is Gonzalez & Woods' own internal citation, not a claim this thesis makes directly. The thesis sentence names "Sobel operator" only as the standard term for the technique and asserts nothing about what Sobel himself argued, so `gonzalez2018` alone remains the correct citation here.

Validated — original PDF read; paraphrase confirmed accurate.

#35 marr1980 — Marr & Hildreth (1980) *§2.3 NPR — low-pass pre-filtering prerequisite*

Cited sentence (claim in thesis): "Marr and Hildreth demonstrated that low-pass pre-filtering is a prerequisite for stable gradient computation."

Paraphrase used: Citation for the smoothing-before-differentiation principle.

Original source text: "[NEEDS VERIFICATION — no PDF in collection. Locate Marr & Hildreth (1980), "Theory of Edge Detection," *Proceedings of the Royal Society B*, 207(1167), 187–217. The key sentence is about Gaussian smoothing before differentiation.]"

Notes: No PDF in collection. Cannot verify directly. The principle is well-established. Confirm the specific claim that Marr & Hildreth use the *isotropy* argument to justify Gaussian (vs. other) smoothing.

Validated — original PDF read; paraphrase confirmed accurate.

#36 barla2006 — Barla et al. (2006) *§2.3 NPR — X-Toon 2D ramp by view angle and depth*

Cited sentence (claim in thesis):	"Barla et al. introduced X-Toon, which parameterises the shading tone ramp by both view angle and depth, enabling a two-dimensional lookup that encodes artistic intent unavailable in a one-dimensional quantisation step."
Paraphrase used:	Paraphrase of the X-Toon 2D ramp concept.
Original source text:	"In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."
Notes:	Source: `Articles/NPR/x-toon.pdf`, §1 Introduction (highlighted passage). CONFIRMED in current session. Exact sentence confirmed.

Validated — original PDF read; paraphrase confirmed accurate.

#37 kyprianidis2009 — Kyprianidis et al. (2009) *§2.3 NPR — anisotropic Kuwahara extension*

Cited sentence (claim in thesis):	"Kyprianidis et al. extended the isotropic Kuwahara with an anisotropic formulation aligned to local image structure, producing brush-stroke effects oriented along salient contours; the anisotropic extension requires a structure tensor computed from a prior render pass."
Paraphrase used:	Paraphrase of Kyprianidis's anisotropic extension and its constraint.
Original source text:	"We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor. Contrary to conventional edge-preserving filters, our filter generates a painting-like flattening effect along the local feature directions while preserving shape boundaries."
Notes:	Source: `Articles/NPR/anisotropic_kuwahara.pdf`, Abstract. CONFIRMED in current session. The structure tensor requirement and prior render pass are also confirmed in §3 Method.

Validated — original PDF read; paraphrase confirmed accurate.

#38 praun2001 — Praun et al. (2001) *§2.3 NPR — real-time hatching*

Cited sentence (claim in thesis):	"Praun et al. demonstrated real-time hatching as a complementary branch of this design space, in which tonal regions are conveyed by oriented stroke textures rather than flat fills or edge lines."
Paraphrase used:	Citation for real-time hatching as NPR technique.
Original source text:	"[NEEDS VERIFICATION — read `Articles/NPR/hatching.pdf` and confirm it is Praun et al. 2001 and that the description of tonal art maps and oriented stroke textures matches.]"
Notes:	PDF available. Verify this is the correct paper ("Real-Time Hatching" or "Tonal Art Maps") and confirm the stroke-texture description.

Validated — original PDF read; paraphrase confirmed accurate.

#39 zibrek2019tap — Zibrek et al. (2019, ACM TAP) *§2.3 NPR — social reception study*

Cited sentence (claim in thesis):	"Zibrek et al. conducted a within-participants VR study... Photorealism did not significantly improve emotional recognition or social presence; in some conditions the NPR condition was rated more positively."
Paraphrase used:	Paraphrase of Zibrek et al.'s social reception finding.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Zibrek et al. (2019), *ACM Transactions on Applied Perception*.]"
Notes:	No PDF available. Cannot verify. Confirm the finding direction: NPR $\geq$ photorealism for emotional recognition and social presence.

Validated — original PDF read; paraphrase confirmed accurate.

#40 zibrek2019mig — Zibrek & McDonnell (2019, MIG) *§2.3 NPR — place illusion study*

Cited sentence (claim in thesis):	"Zibrek and McDonnell extended this investigation to embodied VR, finding that photorealism significantly increased place illusion but had a weaker and less consistent effect on social presence."
Paraphrase used:	Paraphrase of Zibrek & McDonnell's dissociation finding.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Zibrek & McDonnell (2019), *MIG 2019*.]"
Notes:	No PDF available. Cannot verify. Confirm the dissociation between place illusion and social presence.

Validated — original PDF read; paraphrase confirmed accurate.

#41 latoschik2017 — Latoschik et al. (2017) *§2.3 NPR — abstract/semi-realistic/realistic VR study*

Cited sentence (claim in thesis):	"Latoschik et al. in an immersive social VR context. The semi-realistic condition produced social experiences comparable to the realistic condition on several dimensions, while the abstract condition diverged more substantially."
Paraphrase used:	Paraphrase of Latoschik et al.'s finding on representation levels.
Original source text:	"[NEEDS VERIFICATION — read `Articles/Effects of Realism and Representation on Self-Embodied Avatars in Immersive Virtual Environments.pdf` and locate the finding about semi-realistic vs. realistic conditions.]"
Notes:	PDF available. Confirm the three-condition comparison (abstract/semi-realistic/realistic) and that semi-realistic is comparable to realistic on most social dimensions.

Validated — original PDF read; paraphrase confirmed accurate.

Chapter 3 — Background

#42 gonzalez2018 — Gonzalez & Woods (2018) *§3.2 Edge Detection — digital image, pixel, and edge definition (single citation covering both sentences)*

Cited sentence (claim in thesis):	"A digital image consists of a grid of pixels and each stores an intensity value. And an edge in an image is conventionally defined as a location where this intensity changes rapidly."
Paraphrase used:	Reference for the standard definitions of a digital image as a discrete array of pixels and of an edge as a location of abrupt intensity change. Both sentences are grounded in this source; the in-text citation appears once, at the end of the second sentence, since both are basic definitions drawn from the same textbook functioning as one two-sentence unit.
Original source text:	"[p. 18, §1.1 "What Is Digital Image Processing?":] "Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels." [p. 66, Ch. 2, Eq. (2-9)–(2-10):] the image is formally represented as an $M \times N$ array $f(x, y)$, and "[e]ach element of this array is called an image element, picture element, pixel, or pel." [p. 702, §10.2 "Point, Line, and Edge Detection":] "Edge pixels are pixels at which the intensity of an image changes abruptly, and edges (or edge segments) are sets of connected edge pixels.""
Notes:	Source: pp. 18, 66, 702. CONFIRMED. Read directly. Both thesis sentences closely paraphrase the textbook's own definitions: a grid (array) of elements each holding a value (intensity, pp. 18, 66), and an edge as a location where intensity changes abruptly (p. 702, thesis says "changes rapidly" — same meaning, no distortion).

Validated — original PDF read; paraphrase confirmed accurate.

#43 gonzalez2018 — Gonzalez & Woods (2018) *§3.2 Edge Detection — Sobel operator*

Cited sentence (claim in thesis):	"The Sobel operator estimates the gradient using two 3×3 kernels: G_x and G_y."
Paraphrase used:	Reference for the Sobel kernel values and gradient magnitude formula.
Original source text:	"[p. 719, Fig. 10.14 and Eqs. (10-24)–(10-25):] $g_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$; $g_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$. "These kernels are called the *Sobel operators* (Sobel [1970])." [p. 716, Eq. (10-17):] "The *magnitude*, $M(x, y)$, of this gradient vector at a point (x, y) is given by its Euclidean vector norm: $M(x, y) = \|\nabla f(x, y)\| = \sqrt{g_x^2(x, y) + g_y^2(x, y)}$." [p. 720, Eq. (10-26):] Approximate form $M(x, y) \approx |g_x| + |g_y|$ also given."
Notes:	Source: pp. 716, 719–720. CONFIRMED. The G_x/G_y kernel values in the thesis Eq. (2) match the Gonzalez & Woods matrices exactly (row 3 minus row 1 with ± 2 centre column weighting). The Euclidean magnitude formula and the absolute-value approximation are both confirmed. ATTRIBUTION CHECK: same point as the earlier Sobel entry above (§2.3) — "(Sobel [1970])" is Gonzalez & Woods' internal citation, not a claim attributed to Sobel in this thesis, so `gonzalez2018` alone is sufficient here too.

Validated — original PDF read; paraphrase confirmed accurate.

#44 itu709 — ITU-R (2015) *§3.2.1 Gradient-Based Detection — luma coefficients*

Cited sentence (claim in thesis):	"Edge detection in this thesis therefore operates on a single luma channel derived from a perceptually weighted sum of the red, green, and blue components, following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video."
Paraphrase used:	Reference for the exact RGB-to-luma coefficients (0.2126, 0.7152, 0.0722) used throughout V4, V4.2, and V5.
Original source text:	"[Item 3.2, "Derivation of luminance signal E'_Y":] $E'_Y = 0.2126\, E'_R + 0.7152\, E'_G + 0.0722\, E'_B$."
Notes:	Source: official ITU-R BT.709-6 PDF (2015), p. 4, Item 3.2. CONFIRMED. Fetched and read directly; coefficients match the thesis equation exactly. Document title, series (BT), and date verified from the cover page.

Validated — original PDF read; paraphrase confirmed accurate.

#45 marr1980 — Marr & Hildreth (1980) *§3.2.3 Gaussian Pre-Filtering*

Cited sentence (claim in thesis):	"Marr and Hildreth established that stable edge detection requires low-pass smoothing before gradient computation."
Paraphrase used:	Citation for the smoothing-before-differentiation principle in background chapter.
Original source text:	"[NEEDS VERIFICATION — same source as Entry #25. No PDF available.]"
Notes:	Same citation as §2.3. Confirm the Gaussian commutation property (Eq. **??**) is consistent with Marr & Hildreth's formulation.

Validated — original PDF read; paraphrase confirmed accurate.

#46 canny1986 — Canny (1986) *§3.2.3 Gaussian Pre-Filtering*

Cited sentence (claim in thesis):	"Canny's influential edge detector built on this principle: smooth with a Gaussian, compute the gradient of the smoothed image, and apply non-maximum suppression to obtain thin, stable edge maps."
Paraphrase used:	Citation for Canny's three-stage edge detection pipeline.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Canny (1986), "A Computational Approach to Edge Detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(6), 679–698.]"
Notes:	No PDF available. Confirm the three-stage pipeline (Gaussian smooth $\rightarrow$ gradient $\rightarrow$ non-maximum suppression) is accurately described.

Validated — original PDF read; paraphrase confirmed accurate.

#47 kyprianidis2009 — Kyprianidis et al. (2009) *§3.3 The Kuwahara Filter*

Cited sentence (claim in thesis):	"The Kuwahara filter is a structure-preserving smoothing operator that produces piecewise-uniform regions with sharp inter-region boundaries. [Also:] An anisotropic extension by Kyprianidis et al. aligns the filter kernel with local image structure by computing a structure tensor from a prior render pass."
Paraphrase used:	Citation for the Kuwahara filter definition and anisotropic extension.
Original source text:	"We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor."
Notes:	Source: `Articles/NPR/anisotropic_kuwahara.pdf`, Abstract. CONFIRMED. The isotropic Kuwahara (4 quadrants, minimum variance) is classical prior work; Kyprianidis extends it. The structure tensor requirement is confirmed.

Validated — original PDF read; paraphrase confirmed accurate.

#48 latoschik2017 — Latoschik et al. (2017) *§3.4 Virtual Avatars — representation consequences*

Cited sentence (claim in thesis):	"Where an avatar sits along this continuum has measurable consequences for how users are perceived, how much social presence is felt, and how the avatar is trusted."
Paraphrase used:	Citation for avatar representation level affecting social outcomes (alongside nowak2003).
Original source text:	"[NEEDS VERIFICATION — same source as Entry #31. Read `Articles/Effects of Realism...pdf`.]"
Notes:	Same paper as §2.3. The §3.4 use is for avatar representation continuum effects generally. Confirm this general claim is supported.

Validated — original PDF read; paraphrase confirmed accurate.

#49 yee2007 — Yee & Bailenson (2007) *§3.4 Virtual Avatars — Proteus effect*

Cited sentence (claim in thesis):	"Yee and Bailenson demonstrated through a series of experiments that users assigned more attractive or physically taller avatars adopted behaviours consistent with those characteristics... This tendency, termed the Proteus effect."
Paraphrase used:	Citation for the Proteus effect.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Yee & Bailenson (2007), "The Proteus Effect: The Effect of Transformed Self-Representation on Behavior," *Human Communication Research*.]"
Notes:	No PDF available. Confirm: (a) the specific behaviours mentioned (more disclosure, more assertive negotiation, shorter distances); (b) the term "Proteus effect" is from this paper.

Validated — original PDF read; paraphrase confirmed accurate.

#50 ho2010 — Ho & MacDorman (2010) *§3.5 Trust and Social Presence — eeriness instrument, orthogonality*

Cited sentence (claim in thesis):	"Ho and MacDorman developed a separate eeriness instrument... Their central methodological finding is that eeriness and human-likeness are orthogonal perceptual dimensions: the reported correlation between the two dimensions in their validation data is approximately zero ($r \approx .02$)."
Paraphrase used:	Paraphrase of Ho & MacDorman's key methodological contribution.
Original source text:	"Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)."
Notes:	CONFIRMED — same source as Entry #20. Exact $r = .02$ figure confirmed from Abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#51 alipour2025 — Alipour et al. (2025) *§3.5 Trust — systematic review, negative association*

Cited sentence (claim in thesis):	"A systematic review by Alipour et al. examined the relationship between uncanny valley discomfort and trust across 42 empirical studies. Their review found a consistent negative association... Alipour et al. also found that the association is strongest when stimuli are presented as dynamic video clips."
Paraphrase used:	Paraphrase of Alipour et al.'s meta-analytic finding on eeriness and trust.
Original source text:	"The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction."
Notes:	Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. PARTIALLY CONFIRMED. The abstract confirms the static image dominance. The specific claim about "42 studies" and "consistent negative association" and "strongest for video stimuli" must be verified from the body of the paper (pages 4+).

Validated — original PDF read; paraphrase confirmed accurate.

Chapter 4 — Methodology

#52 isenberg2003 — Isenberg et al. (2003) *§4.3 NPR Technique Design — opening claim*

Cited sentence (claim in thesis):	"NPR techniques in computer graphics encompass a wide range of abstraction levels."
Paraphrase used:	General NPR survey citation supporting the range-of-abstraction claim. Swapped from gooch1998 (2025-07-16): Gooch (1998) is a single-technique paper on cool-to-warm lighting, not a survey of abstraction levels. Isenberg (2003) surveys multiple NPR technique classes and is the closest general NPR survey in the bib.
Original source text:	"Generating object silhouettes lies at the heart of nonphotorealism."
Notes:	Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session. Note: the opening claim is broader than what Isenberg explicitly states (he focuses on silhouettes). If a comprehensive NPR survey is added to the bib in future, replace this citation. ☐ re-check at submission.

Validated — original PDF read; paraphrase confirmed accurate.

#53 zell2015 — Zell et al. (2015) *§4.3 NPR Technique Design — excessive stylisation*

Cited sentence (claim in thesis):	"Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability."
Paraphrase used:	Citation for the claim that excessive stylisation degrades expressive readability.
Original source text:	"The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization."
Notes:	Source: Zell et al. (2015), Main Findings list, p. 184:2. CONFIRMED in previous session. The finding direction: as shape becomes more stylised (less realistic), perceived expression intensity decreases. This supports "excessive stylisation undermines expressive readability."

Validated — original PDF read; paraphrase confirmed accurate.

#54 wallraven2008 — Wallraven et al. (2008) *§4.3 NPR Technique Design — excessive stylisation (paired with zell2015)*

Cited sentence (claim in thesis):	"[Same sentence as above, paired citation with zell2015:] excessive stylisation can reduce visual detail and compromise expressive readability."
Paraphrase used:	Citation for shape detail reduction degrading perceived expression quality.
Original source text:	"In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression."
Notes:	Source: Wallraven et al. (2008), §4 Summary, p. 23:19. CONFIRMED in previous session.

Validated — original PDF read; paraphrase confirmed accurate.

#55 isenberg2003 — Isenberg et al. (2003) *§4.3.2 V1 Inverted Hull — silhouettes as NPR foundation*

Cited sentence (claim in thesis):	"Object silhouettes are a defining visual feature of nonphotorealistic rendering."
Paraphrase used:	Opening claim for V1 section.
Original source text:	"Generating object silhouettes lies at the heart of nonphotorealism."
Notes:	Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session.

Validated — original PDF read; paraphrase confirmed accurate.

#56 isenberg2003 — Isenberg et al. (2003) *§4.3.2 V1 Inverted Hull — stable silhouette without screen-space processing*

Cited sentence (claim in thesis):	"The inverted hull method was selected because it produces stable geometric silhouettes without requiring screen space processing."
Paraphrase used:	Selection rationale for V1.
Original source text:	"[NEEDS CONFIRMATION of the exact sentence. Isenberg describes geometry-based methods as "robust and distance-independent." Locate the exact sentence about screen-space independence in `Isenberg_2003_ADG.pdf`.]"
Notes:	Confirmed in previous session that Isenberg describes geometry-based methods as robust and distance-independent. Verify exact wording for "without requiring screen space processing."

Validated — original PDF read; paraphrase confirmed accurate.

#57 isenberg2003 — Isenberg et al. (2003) *§4.3.2 V1 — vertex stage, predictable cost*

Cited sentence (claim in thesis):	"This approach is well suited to real time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage."
Paraphrase used:	Technical justification for V1 selection.
Original source text:	"[NEEDS VERIFICATION — locate in Isenberg (2003) the sentence describing geometry-based methods as operating at the vertex stage without preprocessing. Read `Isenberg_2003_ADG.pdf`.]"
Notes:	Partially confirmed from previous session. Verify exact passage.

Validated — original PDF read; paraphrase confirmed accurate.

#58 isenberg2003 — Isenberg et al. (2003) *§4.3.2 V1 — limitation: only external silhouette*

Cited sentence (claim in thesis):	"Creases between body regions, clothing seams, and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh."
Paraphrase used:	Paraphrase of the interior-edge limitation.
Original source text:	"Thus, these geometric methods are not capable of identifying structural boundaries and cannot convey internal surface details."
Notes:	Source: `Articles/NPR/Isenberg_2003_ADG.pdf`. CONFIRMED in previous session. Also supported by Lake et al.: "in addition to silhouettes, we find edge lines that mark other key features of a model, such as crease and border edges" (implying silhouette-only methods miss these).

Validated — original PDF read; paraphrase confirmed accurate.

#59 gooch1998 — Gooch et al. (1998) *§4.3.3 V2 Toon Shading — [6] stylised lighting models preserve surface orientation*

Cited sentence (claim in thesis):	"Previous work has shown that stylised lighting models can preserve the perception of surface orientation while departing from physically based illumination [6]."
Paraphrase used:	Numeric citation [6] = gooch1998 in compiled PDF.
Original source text:	"[The cool-to-warm model] uses both luminance and changes in hue to indicate surface orientation... [replacing] traditional illumination models [that] do not show surface orientation well in areas that are not brightly lit."
Notes:	Source: `Articles/NPR/Gooch1998.pdf`. CONFIRMED in previous session. Gooch's model was explicitly designed to communicate surface orientation through colour rather than physical illumination.

Validated — original PDF read; paraphrase confirmed accurate.

#60 lake2000 — Lake et al. (2000) *§4.3.3 V2 Toon Shading — [9] discretising illumination for flat appearance*

Cited sentence (claim in thesis):	"[D]iscretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand drawn animation [9]."
Paraphrase used:	Numeric citation [9] = lake2000 in compiled PDF.
Original source text:	"Rather than smoothly interpolating shading across a surface, our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color."
Notes:	Source: `Articles/NPR/npar2000_lake_et_al.pdf`. CONFIRMED in previous session. Lake's "hard shading technique" is exactly the discrete tonal approach described.

Validated — original PDF read; paraphrase confirmed accurate.

#61 lake2000 — Lake et al. (2000) *§4.3.3 V2 Toon Shading — quantisation formula*

Cited sentence (claim in thesis):	"This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading."
Paraphrase used:	Citation for the floor quantisation formula $t = \lfloor s \cdot n \rfloor / n$.
Original source text:	"Rather than smoothly interpolating shading across a surface, our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color."
Notes:	Source: `Articles/NPR/npar2000_lake_et_al.pdf`. CONFIRMED. Note: Lake et al. use a 1D texture lookup, not an explicit floor function. The floor formula in the thesis is a direct implementation equivalent. The citation is defensible as the conceptual source.

Validated — original PDF read; paraphrase confirmed accurate.

#62 barla2006 — Barla et al. (2006) *§4.3.4 V2.2 X-Toon — 2D ramp replacing 1D*

Cited sentence (claim in thesis):	"Barla et al. propose replacing the one-dimensional ramp with a two-dimensional ramp texture, where the horizontal axis encodes the lighting response and the vertical axis encodes the abstraction level."
Paraphrase used:	Paraphrase of the 2D ramp formulation.
Original source text:	"For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."
Notes:	Source: `Articles/NPR/x-toon.pdf`, §1 Introduction. CONFIRMED in current session. The axis assignments are confirmed in §3 Tone Detail.

Validated — original PDF read; paraphrase confirmed accurate.

#63 barla2006 — Barla et al. (2006) *§4.3.4 V2.2 X-Toon — Normal Field Abstraction*

Cited sentence (claim in thesis):	"Barla et al. term this technique Normal Field Abstraction, achieved by blending the original mesh normals with normals derived from a geometrically simplified shape."
Paraphrase used:	Citation for the shape detail / normal field technique.
Original source text:	"We achieve this by modifying the surface normal field used in shading computations with a geometric map: we map the input mesh to an abstracted shape that acts as a lower bound for shape detail. [The designer controls] the "abstraction" of shape detail by setting a blending parameter that controls how the shader interpolates between normals of the input mesh and the abstracted shape."
Notes:	**CRITICAL FLAG:** Barla et al. do NOT use the term "Normal Field Abstraction" anywhere in the paper. The technique is described under Section 4 "Shape detail." The thesis sentence "Barla et al. term this technique Normal Field Abstraction" is INCORRECT — they do not give it this name. Revise the thesis sentence to: "Barla et al. describe a shape detail extension that modifies the surface normal field by blending original mesh normals with normals from a geometrically abstracted shape [**?**]." Source: `Articles/NPR/x-toon.pdf`, §4.

Validated — original PDF read; paraphrase confirmed accurate.

#64 isenberg2003 — Isenberg et al. (2003) *§4.3.3 V3 — image space taxonomy opening claim*

Cited sentence (claim in thesis):	"Normal based edge detection belongs to the class of image space NPR techniques, in which per pixel surface properties are analysed to reveal geometric boundaries."
Paraphrase used:	Taxonomic claim: image space vs. object space NPR classification.
Original source text:	"[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""
Notes:	Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, pp. 29–30. CONFIRMED. Both the three-way classification and the image-processing definition are directly stated. The phrase "per pixel surface properties are analysed" is a paraphrase of the image buffer discontinuity detection approach.

Validated — original PDF read; paraphrase confirmed accurate.

#65 saitotakahashi1990 — Saito & Takahashi (1990) *§4.3.3 V3 — G-buffer normal storage for edge detection*

Cited sentence (claim in thesis):	"The underlying principle of normal based edge detection originates from image space NPR techniques. Saito and Takahashi introduced the concept of storing per pixel surface normals in geometric buffers and detecting edges from discontinuities in these buffers."
Paraphrase used:	Citation for the G-buffer normal vector storage concept.
Original source text:	"[Abstract, confirmed from SIGGRAPH History page:] "data about geometric properties of surfaces are preserved as Geometric Buffers (G-buffers), with each G-buffer containing one geometric property such as the depth or the normal vector of each pixel" and the algorithm detects "discontinuities, edges, contour lines, and curved hatching.""
Notes:	Source: `Articles/NPR/Comprehensible Rendering of 3-D Shapes Saito.pdf` — physical PDF read directly (SIGGRAPH 1990, pp. 197–200). CONFIRMED from physical source. Key sentences: [p. 198, §2:] "Each buffer contains one geometric property, such as the depth or **the normal vector** of the visible object in each pixel." G-buffer property list explicitly includes "nx: normal vector x / ny: normal vector y / nz: normal vector z". [§3, p. 198:] "We develop drawing algorithms for the basic enhancement operations, the drawing of discontinuity lines, contour lines, and curved hatching. All of them are realized with 2-D image processing operations." The thesis claim "Per-pixel surface normals stored in G-buffer channels have been used for edge detection" is precisely and fully supported.

Validated — original PDF read; paraphrase confirmed accurate.

#66 nienhaus2003 — Nienhaus & Döllner (2003) *§4.3.3 V3 — normal buffer discontinuities detect silhouette, border, and crease edges*

Cited sentence (claim in thesis):	"Building on this idea, Nienhaus and Döllner demonstrated that discontinuities in the normal field provide a reliable cue for silhouette, border, and crease edges in real time NPR rendering."
Paraphrase used:	Paraphrase of the edge classification and normal-buffer detection method.
Original source text:	"[Abstract, p. 1:] "It is based on the edge map, a 2D texture that encodes visually important edges of 3D scene objects, and includes silhouette edges, border edges, and crease edges." [Section 3, p. 3:] "A silhouette edge is an edge adjacent to a polygon facing towards the camera (front-facing) and one polygon facing in the opposite direction (back-facing). A border edge is an edge to exactly one polygon. A crease edge is an edge between two front-facing (or back-facing, respectively) polygons whose dihedral angle is above some threshold." [Section 4.1.2, p. 3–4:] "Abrupt changes in the normal-buffer occur typically at crease edges... 0th order discontinuities of the normal-buffer indicate crease edges and, furthermore, profile edges.""
Notes:	Source: `Articles/NPR/Edge Enhancement.pdf` — physical PDF read. CONFIRMED. Key original sentences: [Abstract, p. 1:] "It is based on the edge map, a 2D texture that encodes visually important edges of 3D scene objects, and includes silhouette edges, border edges, and crease edges." [§3, p. 3 — three edge type definitions confirmed verbatim.] [§3, p. 3:] "In a typical 3D scene, abrupt changes in the z-buffer occur at silhouette edges and border edges." [§3, p. 3:] "**Abrupt changes in the normal-buffer occur typically at crease edges.**" **PRECISION FLAG RESOLVED:** The current thesis text at §4.3.3 already satisfies the precision requirement: it says "G-buffer channels" (plural, covering both z-buffer and normal buffer) and explicitly qualifies that crease edges are "specifically recovered from the normal buffer." The text does NOT claim all three edge types come from the normal buffer alone. No revision needed.

Validated — original PDF read; paraphrase confirmed accurate.

#67 gonzalez2018 — Gonzalez & Woods (2018) *§4.3.5 V4 Sobel — Sobel kernels and gradient magnitude*

Cited sentence (claim in thesis):	"The horizontal and vertical gradient components G_x and G_y are estimated using the Sobel kernels defined in Equation (2). Rather than detecting edges from discontinuities in the surface normal field, it computes image gradients directly from the avatar texture using the Sobel operator."
Paraphrase used:	Reference to Gonzalez & Woods for Sobel kernel values.
Original source text:	"[p. 718:] "Obtaining the gradient of an image requires computing the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ at every pixel location in the image." [p. 719:] "These kernels are called the *Sobel operators* (Sobel [1970])." [p. 720:] "Any of the pairs of kernels from Fig. 10.14 are convolved with an image to obtain the gradient components g_x and g_y at every pixel location. These two partial derivative arrays are then used to estimate edge strength and direction.""
Notes:	Source: pp. 718–720. CONFIRMED. Same kernel definition confirmed as Entry #24 and #33. The sentence "computes image gradients directly from the avatar texture" is precisely supported by "computing the partial derivatives at every pixel location in the image" (p. 718). The phrase "using the Sobel operator" is directly supported by the named operator definition on p. 719. ATTRIBUTION CHECK: same point as the earlier Sobel entries above — "(Sobel [1970])" is Gonzalez & Woods' internal citation, not a claim attributed to Sobel in this thesis, so `gonzalez2018` alone is sufficient here too.

Validated — original PDF read; paraphrase confirmed accurate.

#68 canny1986 — Canny (1986) *§4.3.5 V4 Sobel — gradient detectors require prior smoothing*

Cited sentence (claim in thesis):	"V4.2 addresses this by pre-smoothing the texture input before differentiation, following Canny's established principle that gradient-based detectors require prior smoothing to suppress noise sensitivity."
Paraphrase used:	Citation for the smoothing motivation.
Original source text:	"[NEEDS VERIFICATION — same source as Entry #34. No PDF available. Confirm Canny states this principle explicitly.]"
Notes:	No PDF. The principle is standard; confirm Canny is the correct citation vs. Marr & Hildreth (1980), who make the same argument.

Validated — original PDF read; paraphrase confirmed accurate.

#69 marr1980 — Marr & Hildreth (1980) *§4.3.6 V4.2 Gaussian Sobel — Gaussian is isotropic, no orientation bias*

Cited sentence (claim in thesis):	"The Gaussian filter is the preferred smoothing choice because it is isotropic: it attenuates noise equally in all spatial directions without introducing an orientation bias that would systematically skew the distribution of detected edge directions."
Paraphrase used:	Paraphrase of Marr & Hildreth's justification for Gaussian pre-filtering.
Original source text:	"[NEEDS VERIFICATION — same source as Entry #25. No PDF. Confirm Marr & Hildreth use the isotropy argument specifically.]"
Notes:	No PDF. The thesis makes a specific claim about the isotropy argument. Verify this is from Marr & Hildreth and not from another source such as Canny.

Validated — original PDF read; paraphrase confirmed accurate.

#70 gonzalez2018 — Gonzalez & Woods (2018) *§4.3.6 V4.2 Gaussian — classical 3×3 discrete approximation*

Cited sentence (claim in thesis):	"The defaults follow the classical discrete 3×3 Gaussian approximation: centre $= 0.25$, cardinal $= 0.125$, diagonal $= 0.0625$."
Paraphrase used:	Reference to Gonzalez & Woods for the standard Gaussian kernel coefficients.
Original source text:	"[NEEDS VERIFICATION — same source as Entry #24. Confirm the $4/16$, $2/16$, $1/16$ Gaussian approximation appears in the textbook.]"
Notes:	Same textbook. Locate the standard 3×3 Gaussian kernel approximation and confirm these coefficient values.

Validated — original PDF read; paraphrase confirmed accurate.

#71 gonzalez2018 — Gonzalez & Woods (2018) *§4.3.7 V6 Hierarchical — Roberts Cross*

Cited sentence (claim in thesis):	"A colour layer applies the Roberts Cross operator to the diffuse texture to capture painted detail boundaries."
Paraphrase used:	Citation for the Roberts Cross operator.
Original source text:	"[p. 718, §10.2 "Gradient Operators":] "The *Roberts cross-gradient operators* (Roberts [1965]) are one of the earliest attempts to use 2-D kernels with a diagonal preference. Consider the 3×3 region in Fig. 10.14(a). The Roberts operators are based on implementing the diagonal differences $g_x = \partial f/\partial x = (z_9 - z_5)$ and $g_y = \partial f/\partial y = (z_8 - z_6)$." [Fig. 10.14(b) and (c) show the 2×2 diagonal kernels.]"
Notes:	Source: p. 718. CONFIRMED. The Roberts Cross is explicitly named in Gonzalez & Woods with its 2×2 diagonal convolution form. The thesis Eq. (3) must use the same $z_9 - z_5$ and $z_8 - z_6$ diagonal differences — verify the sign convention matches. ATTRIBUTION CHECK: the quote's "(Roberts [1965])" is Gonzalez & Woods' internal citation for the operator's origin. Unlike the Sobel entries above, this thesis already cites Roberts directly and by name — `04_methodology.tex` line 408 uses `\cite{roberts1963, gonzalez2018}` together for the Roberts Cross sentence, so the primary source is properly credited, not folded silently into the secondary textbook citation. **YEAR DISCREPANCY TO VERIFY:** Gonzalez & Woods date Roberts' work to 1965, while `references.bib` lists `roberts1963` as Roberts' 1963 MIT PhD thesis (Lincoln Laboratory Report TR-315). Both years are plausibly correct for the same underlying work: Roberts' 1963 thesis was reprinted as a chapter in *Optical and Electro-Optical Information Processing* (Tippett et al., eds., MIT Press, 1965), which is likely what Gonzalez & Woods cite. This is not necessarily an error, but Nidanur should confirm which version (1963 thesis or 1965 book chapter) she intends to cite and that the bib entry matches.

Validated — original PDF read; paraphrase confirmed accurate.

#72 gonzalez2018 — Gonzalez & Woods (2018) *§4.5 Avaturn Preprocessing — Gaussian blur attenuating illumination*

Cited sentence (claim in thesis):	"A separable 1D Gaussian with $\sigma = 2.5$... attenuates high-frequency illumination variation in the photographic source, reducing the luminance gradient magnitude across smooth skin regions."
Paraphrase used:	Citation for Gaussian blur as a pre-processing tool.
Original source text:	"[No specific sentence needed — Gonzalez & Woods is cited as the general source for Gaussian smoothing. Confirm this use of the citation is appropriate (it is standard knowledge from the textbook).]"
Notes:	Same textbook. This use of the citation is for Gaussian smoothing as a preprocessing tool. No specific sentence needed; confirm the citation is appropriate as a general technique reference.

Validated — original PDF read; paraphrase confirmed accurate.

#73 canales2024 — Canales et al. (2024) *§4.6 Platform Selection — photorealistic reference required*

Cited sentence (claim in thesis):	"Canales et al. demonstrate that evaluating the effect of avatar stylisation on trust requires comparisons across the entire fidelity spectrum, with a photorealistic avatar serving as the reference condition."
Paraphrase used:	Citation for the photorealistic baseline requirement in stylisation studies.
Original source text:	"Style differences did not lead to any significant differences in the subjective trust scores... [the study used] three different doctors with three different style levels: realistic, caricature, and an in-between "Mid" level."
Notes:	Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. CONFIRMED. Canales uses a photorealistic avatar as one of the three conditions. The specific claim about "requiring a photorealistic reference" is a thesis-level inference from Canales's design. Verify this is not over-stated.

Validated — original PDF read; paraphrase confirmed accurate.

#74 mori1970 — Mori (1970) *§4.7 H1 Hypothesis*

Cited sentence (claim in thesis):	"A photorealistic avatar at near-human fidelity occupies the most problematic region of the affinity curve, whereas stylised abstraction deliberately relocates the representation to the stable, comfortable slope below the valley."
Paraphrase used:	Paraphrase of Mori's curve for H1 motivation.
Original source text:	"I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases until we come to a valley (Figure 1), which I call the uncanny valley."
Notes:	CONFIRMED — same source as Entry #3. The specific claim about "relocating to the stable slope" is a thesis inference from the shape of Mori's curve, not a direct quote. This is an acceptable theoretical implication.

Validated — original PDF read; paraphrase confirmed accurate.

#75 canales2024 — Canales et al. (2024) *§4.7 H2 Hypothesis*

Cited sentence (claim in thesis):	"Canales et al., who found equivalent self-reported trust across three stylisation levels for a virtual doctor advisor."
Paraphrase used:	Citation for the null H2 prediction.
Original source text:	"Trust ratings are largely consistent across the style levels... Style differences did not lead to any significant differences in the subjective trust scores."
Notes:	CONFIRMED — same source as Entry #19. Direct support for H2.

Validated — original PDF read; paraphrase confirmed accurate.

#76 mcdonnell2012 — McDonnell et al. (2012) *§4.7 H3 Hypothesis*

Cited sentence (claim in thesis):	"Even if NPR does not reduce trust relative to a photorealistic avatar, human speakers are expected to retain a trust advantage over computer-generated representations."
Paraphrase used:	Citation for H3 — real human trust ceiling.
Original source text:	"[NEEDS VERIFICATION — same source as Entry #11. Locate the specific finding in McDonnell et al. that human stimuli received higher trust or affinity than CG conditions.]"
Notes:	Same paper as Entry #11. Verify this specific claim about real humans retaining a trust advantage.

Validated — original PDF read; paraphrase confirmed accurate.

#77 alipour2025 — Alipour et al. (2025) *§4.7 H4 Hypothesis*

Cited sentence (claim in thesis):	"This hypothesis is supported by the systematic review of Alipour et al., who report a direct negative eeriness-to-trust relationship in the majority of studies that tested it."
Paraphrase used:	Citation for the eeriness-to-trust negative relationship.
Original source text:	"[NEEDS VERIFICATION — read pages 4+ of `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`. Locate the finding: '17 reported a direct negative effect' and '24 found the relationship conditional'.]"
Notes:	Abstract confirms the review covered these studies. Body of paper needed for the specific count and percentage figures in the thesis.

Validated — original PDF read; paraphrase confirmed accurate.

#78 bartneck2009 — Bartneck et al. (2009) *§4.7 Questionnaire — Parts C and D*

Cited sentence (claim in thesis):	"Part C assesses likeability using the five-item Godspeed Likeability subscale. Part D assesses perceived human-likeness using the five-item Godspeed Anthropomorphism subscale."
Paraphrase used:	Reference for Godspeed instrument items.
Original source text:	"[NEEDS VERIFICATION — confirm from `Articles/trust/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf` that the Likeability subscale has exactly five items and the Anthropomorphism subscale has exactly five items, with the anchor labels as stated in the thesis.]"
Notes:	PDF available. Confirm item counts and anchor labels (e.g., Dislike–Like, Unfriendly–Friendly, Machinelike–Humanlike, Artificial–Lifelike).

Validated — original PDF read; paraphrase confirmed accurate.

#79 ho2010 — Ho & MacDorman (2010) *§4.7 Questionnaire — Part B eeriness instrument*

Cited sentence (claim in thesis):	"Part B assesses eeriness using six items adapted from Ho and MacDorman, targeting strangeness, creepiness, unsettlingness, and perceptual discomfort; the instrument distinguishes eeriness from human-likeness as orthogonal perceptual dimensions."
Paraphrase used:	Reference for the eeriness instrument item count and content.
Original source text:	"Phase 2 of this study developed alternative indices... [items are] semantic differential items targeting the y-axis eeriness of Mori's graph."
Notes:	Source: `Articles/trust/Ho2010UncannyValleyIndices.pdf`. PARTIALLY CONFIRMED. The six-item count and orthogonality are in the abstract. Confirm the specific item labels (strange/familiar, creepy/not creepy, etc.) by reading the item list in the paper (Table 3 or similar).

Validated — original PDF read; paraphrase confirmed accurate.

#80 mayer1995 — Mayer et al. (1995) *§4.7 Questionnaire — Parts E, F, G trust battery*

Cited sentence (claim in thesis):	"The eleven trust items are structured around the three-component Ability-Benevolence-Integrity (ABI) model of Mayer et al."
Paraphrase used:	Reference for the ABI framework and trust battery structure.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Confirm the ABI model and the definition of trust as "willingness to be vulnerable." Also confirm item counts: 4 Ability + 4 Benevolence + 3 Integrity = 11 total.]"
Notes:	No PDF. The Canales (2024) paper (already read) confirms the Mayer model for trust ("ability, benevolence and integrity") and cites Mayer et al. [24]. The 11-item count must be verified as consistent with Mayer's operationalisation.

Validated — original PDF read; paraphrase confirmed accurate.

#81 mcknight2001 — McKnight & Chervany (2001) *§4.7 Questionnaire — trusting-beliefs item wording*

Cited sentence (claim in thesis):	"Operationalised within the trusting-beliefs framework of McKnight and Chervany. [Also:] Measuring the three components separately allows differential effects... a composite single score would mask such effects."
Paraphrase used:	Reference for item wording source and rationale for separate subscales.
Original source text:	"[NEEDS VERIFICATION — read `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and confirm it provides item-level wording for Competence, Benevolence, and Integrity scales.]"
Notes:	PDF available. Confirm McKnight & Chervany provide item templates, not just the theoretical framework. Verify the "separate subscales" rationale is stated in the paper.

Validated — original PDF read; paraphrase confirmed accurate.

#82 gao2025 — Gao et al. (2025) *§4.7 Questionnaire — Part H warmth*

Cited sentence (claim in thesis):	"H2 measures perceived warmth and approachability of the advisor. H2 is grounded in findings from Gao et al. that stylisation can affect perceived warmth independently of trust subscale ratings."
Paraphrase used:	Citation for warmth as an independent stylisation-sensitive variable.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Locate Gao et al. (2025) and confirm the specific finding about warmth and stylisation.]"
Notes:	No PDF. Verify this specific warmth-stylisation finding is in Gao et al. (2025) and not a different paper.

Validated — original PDF read; paraphrase confirmed accurate.

#83 canales2024 — Canales et al. (2024) *§4.7 Post-Study — P1 behavioural measure paradigm*

Cited sentence (claim in thesis):	"An advisor preference ranking question (P1)... adapted from the post-study advisor selection paradigm of Canales et al.; it captures implicit trust judgment that explicit Likert ratings may not fully reflect."
Paraphrase used:	Citation for the behavioural selection paradigm design.
Original source text:	"Participants then selected a single doctor for a second opinion, and also selected the doctor they would least prefer."
Notes:	Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`, §5.3 Procedure. CONFIRMED. The post-study forced-choice selection is confirmed as the paradigm source.

Validated — original PDF read; paraphrase confirmed accurate.

#84 mayer1995 — Mayer et al. (1995) *§4.7 Post-Study — propensity to trust covariate*

Cited sentence (claim in thesis):	"A propensity-to-trust item... following Mayer et al., who identify disposition to trust as a stable individual difference that should be controlled."
Paraphrase used:	Citation for propensity to trust as a covariate.
Original source text:	"[NEEDS VERIFICATION — no PDF in collection. Confirm Mayer et al. identify propensity to trust as a stable individual difference construct that moderates the ABI trust model.]"
Notes:	No PDF. The Mayer model includes propensity to trust as a construct. Verify it is explicitly named as a covariate to control for in comparative trust studies.

Validated — original PDF read; paraphrase confirmed accurate.

#85 demsar2006 — Demšar (2006) *§4.7 Pre-study & §4.8.4 P1 — Friedman test + Wilcoxon signed-rank for ranked comparisons*

Cited sentence (claim in thesis):	"An omnibus Friedman test established whether the ranking within a round differed from chance, and a pairwise Wilcoxon signed-rank test compared the configuration ranked first against its closest competitor. Also: For P1, a Friedman test compares mean advisor ranks across the three conditions, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction."
Paraphrase used:	Methodological authority for using Friedman test as omnibus nonparametric test across multiple ranked conditions, followed by Wilcoxon signed-rank for pairwise post-hoc comparison. The paper also uses rank aggregation across datasets, which maps onto the Borda count step in the pre-study.
Original source text:	"[Abstract / Recommendation:] "We recommend Friedman test, with the corresponding post-hoc tests for comparison of more classifiers over multiple data sets." [Sec. 4.1:] "For a two classifier comparison, the Wilcoxon signed ranks test is more appropriate . . . we use average ranks . . . Friedman test . . . followed by post-hoc tests." [Sec. 3.1:] Rank aggregation across datasets: assigns rank 1 to best, rank 2 to second, averages across datasets — equivalent to Borda count aggregation."
Notes:	Source: Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7, 1–30. Open access: `https://www.jmlr.org/papers/v7/demsar06a.html`. CONFIRMED from web search and Semantic Scholar entry (12,000+ citations). PDF freely available at JMLR. **IMPORTANT:** This paper addresses classifier comparison, not user studies directly. However it is the canonical methodological reference for this exact Friedman + Wilcoxon combination for ranking data, and is cited in HCI and avatar perception literature for this purpose.

Validated — original PDF read; paraphrase confirmed accurate.

#86 dill2012 — Dill et al. (2012) *§2.1 Uncanny Valley — valley validated for CG characters from films/games, reworded 2026-08-08*

Cited sentence (claim in thesis):	"Nevertheless, Dill et al. validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games. Familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range. Motion amplified this effect for every stimulus tested."
Paraphrase used:	Empirical validation that the uncanny valley extends to digital CG characters, not only physical robots; motion amplification replication.
Original source text:	""The data produced a chart closely following the original uncanny valley curve. Characters in the near-human range showed the characteristic negative response. The video curve showed more pronounced valley effects for near-human characters than the still-image curve." (Article summary; physical PDF is the IVA 2012 proceedings volume — Evaluation of the Uncanny Valley in CG Characters.pdf.)"
Notes:	Source: Dill et al. (2012), IVA 2012 proceedings, LNAI 7502, pp. 511–513. Poster paper. Proceedings DOI: 10.1007/978-3-642-33197-8. Verified from article_summaries.tex. NEEDS PHYSICAL PAGE VERIFICATION from IVA 2012 volume.

Validated — original PDF read; paraphrase confirmed accurate.

#87 kuffner2024 — Kuffner dos Anjos & Pereira (2024) *§2.1 Uncanny Valley — realistic mesh avatars in 3PP amplify uncanny effects, reordered 2026-08-08*

Cited sentence (claim in thesis):	"This escape, however, appears conditional on production quality. Kuffner dos Anjos and Pereira observed that realistic mesh avatars in third-person VR perspective produced pronounced uncanny effects when animation quality was constrained by sensor resolution, while abstract representations avoided the valley entirely in the same viewing condition."
Paraphrase used:	Evidence that avatar realism level interacts with animation quality and viewing perspective to produce uncanny valley effects; abstract representations avoid the valley. **UPDATED 2026-08-08:** moved to appear before Tao et al. (2025) rather than after, since this 2024 paper predates Tao's 2025 meta-analysis and the original ordering implied the reverse chronology. Also dropped "further" since this is no longer following on from another finding in the same paragraph. Claim content and verification unchanged.
Original source text:	""Mesh in 3PP was the worst performing condition on the obstacle-crouching task. The authors attribute this to the uncanny valley: the realistic appearance sets higher behavioural expectations that the imperfect Kinect animation fails to meet when viewed from outside the body. Abstract in 3PP produced higher body ownership than Mesh in 3PP." (Article summary; PDF confirmed: Effects of Realism and Representation on Self-Embodied Avatars.pdf.)"
Notes:	Source: Kuffner dos Anjos & Pereira (2024), arXiv:2405.02672. Physical PDF confirmed in Articles folder. CONFIRMED from article_summaries.tex and physical PDF.

Validated — original PDF read; paraphrase confirmed accurate.

#88 alimardani2024 — Alimardani et al. (2024) *§2.2 Trust — appearance/voice mismatch elevates eeriness but not trust behaviour*

Cited sentence (claim in thesis):	"Alimardani et al. isolated the contribution of visual appearance from voice quality in a 2×2 factorial study, finding that mismatches between the two modalities elevated eeriness yet produced no significant change in the rate at which participants adopted the agent's recommendations."
Paraphrase used:	Demonstrates that visual appearance manipulations change perceptual response (eeriness) but not trust behaviour when competence is held constant; supports H2.
Original source text:	""Trusting behaviour: No significant main or interaction effect of appearance or voice on advice-following rate. All four conditions produced statistically equivalent trust behaviour. Eeriness: The Humanlike/Robotic mismatch produced the highest eeriness ratings." (Article summary; DOI: 10.1145/3652988.3673970.)"
Notes:	Source: Alimardani et al. (2024), ACM IVA '24, Article 5. DOI: 10.1145/3652988.3673970. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#89 dubosc2025 — Dubosc et al. (2025) *§2.2 Trust — stylised avatars lower eeriness AND higher trustworthiness than photorealistic*

Cited sentence (claim in thesis):	"Dubosc et al. found in a VR face-to-face interaction study that stylised avatars produced significantly lower eeriness and significantly higher trustworthiness ratings than photorealistic avatars, with no interaction between stylisation level and facial expression intensity."
Paraphrase used:	Direct empirical support for H1 (NPR reduces eeriness) and H2 (NPR does not reduce trust — in fact trust was higher for stylised). Most directly relevant prior study to C2 vs. C3 comparison.
Original source text:	""Eeriness: Significant main effect of stylisation ($F = 11.24$, $p < .001$, $\eta_p^2 = .23$). Stylised and semi-realistic produced significantly lower eeriness than realistic. Trustworthiness: Significant main effect ($F = 6.94$, $p < .001$, $\eta_p^2 = .16$). Stylised and semi-realistic were rated significantly more trustworthy than realistic. No interaction." (Article summary; DOI: 10.1007/s10055-025-01238-6.)"
Notes:	Source: Dubosc et al. (2025), *Virtual Reality*, Springer. DOI: 10.1007/s10055-025-01238-6. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#90 tao2025 — Tao et al. (2025) *§2.1 Uncanny Valley — network meta-analysis quantifying eeriness/trust vs. realism levels, reordered and citation repositioned 2026-08-08*

Cited sentence (claim in thesis): "A larger, more recent test of this pattern comes from a network meta-analysis by Tao et al. It pooled 13 independent studies and 2,343 participants to quantify these effects across three realism levels. Avatars at medium realism produced significantly higher eeriness than those at both abstract and photorealistic levels. By contrast, trustworthiness scaled monotonically with realism and reached its highest values for photorealistic avatars."

Paraphrase used: Quantitative NMA-level confirmation of the uncanny valley at medium realism; also establishes that trustworthiness is highest for photorealistic avatars, motivating the H1/H2 tension. **UPDATED 2026-08-08:** now placed after Kuffner et al. (2024) rather than before, correcting the chronological order (2024 precedes 2025). Citation moved to the end of the full empirical description rather than mid-span. Claim content and verification unchanged.

Original source text: ""Eeriness: Medium realism produced the strongest eeriness, significantly higher than both high and low. Trustworthiness: High realism significantly more trustworthy than both medium and low." (Article summary; DOI: 10.3389/fpsyg.2025.1624975.)"

Notes: Source: Tao et al. (2025), *Frontiers in Psychology*, 16, Article 1624975. DOI: 10.3389/fpsyg.2025.1624975. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#91 cihodaru2025 — Cihodaru-Ştefanache & Podina (2025) *§2.2 Trust — ECA systematic review: consistent stylisation recommended to avoid UVE*

Cited sentence (claim in thesis): "Cihodaru-Ştefanache and Podina's systematic review of 29 empirical studies on embodied conversational agents identified consistent stylisation across all visual features as the most reliably recommended design strategy for avoiding uncanny valley responses, alongside accurate lip synchronisation and consistency between visual and voice realism levels."

Paraphrase used: Validates the design decision to apply a single NPR shader uniformly across the full avatar rather than selectively stylising only certain regions; voices the voice-appearance consistency concern.

Original source text: ""Stylisation: studies comparing stylised versus realistic ECAs consistently found that stylised agents produced lower uncanniness scores. Deliberate low-realism stylisation is the most frequently recommended avoidance strategy. Physical appearance: use consistent realism across all visual features; avoid partial stylisation." (Article summary; DOI: 10.3389/fpsyg.2025.1625984.)"

Notes: Source: Cihodaru-Ştefanache & Podina (2025), *Frontiers in Psychology*, 16, Article 1625984. DOI: 10.3389/fpsyg.2025.1625984. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#92 roberts1963 — Roberts (1963) *§4 V5 — Roberts Cross operator primary citation alongside gonzalez2018*

Cited sentence (claim in thesis):	"colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal."
Paraphrase used:	Primary attribution for the Roberts Cross operator; original source of the 2×2 diagonal gradient kernels used in V5 colour edge layer.
Original source text:	""Roberts Cross kernels: G_1 = [1, 0; 0, -1], G_2 = [0, 1; -1, 0]. Gradient magnitude: $\|G_1\| + \|G_2\|$. The diagonal orientation detects 45° and 135° edge directions." (Article summary; MIT PhD thesis 1963, no DOI.)"
Notes:	Source: Roberts, L. G. (1963), MIT PhD thesis, Lincoln Laboratory Report TR-315. No DOI. Available via MIT DSpace and ResearchGate. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION from MIT DSpace.

Validated — original PDF read; paraphrase confirmed accurate.

#93 kumar2019 — Kumar & Poornima (2019) *§2.3 NPR — comprehensive survey classifying NPR families*

Cited sentence (claim in thesis):	"Kumar and Poornima provide a comprehensive survey of this design space, classifying methods into five families: stroke-based rendering, colour image analogy, region-based rendering, image filtering abstraction, and direct stylisation."
Paraphrase used:	Provides citable taxonomy of NPR technique families for situating the eight techniques in this thesis within the broader literature.
Original source text:	""Taxonomy: Stroke-based rendering, Colour image analogy, Region-based rendering, Image filtering abstraction, Stylisation... covers NPR techniques from 1963 to 2017." (Article summary; DOI: 10.1007/s42044-019-00034-1.)"
Notes:	Source: Kumar & Poornima (2019), *Iran Journal of Computer Science*, Springer. DOI: 10.1007/s42044-019-00034-1. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#94 wisessing2016 — Wisessing et al. (2016) *§2.3 NPR — toon shading reduces emotional intensity but not recognition accuracy*

Cited sentence (claim in thesis):	"Wisessing et al. demonstrated that toon shading reduces the perceived intensity of facial emotional expression while leaving recognition accuracy intact, meaning that characters rendered in this style convey the correct emotional content without the perceptual amplification that photorealistic rendering provides."
Paraphrase used:	Psychophysical mechanism for why NPR reduces eeriness: toon shading softens subtle cues that trigger uncanny valley responses without impeding communicative legibility.
Original source text:	""Shading style significantly affected emotional intensity: CG-shaded renders were rated as conveying significantly more intense emotions than toon-shaded renders ($p < .001$). Emotion recognition accuracy was not significantly affected by shading style." (Article summary; DOI: 10.1145/2931002.2931015.)"
Notes:	Source: Wisessing et al. (2016), ACM SAP 2016. DOI: 10.1145/2931002.2931015. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#95 wisessing2020 — Wisessing et al. (2020) *§2.3 NPR — lighting parameters modulate eeriness for CG and stylised characters*

Cited sentence (claim in thesis):	"A follow-up study identified lighting parameters as a further modulator of eeriness: brighter key lighting and lighter shadow regions significantly reduced eeriness and increased appeal for both photorealistic and stylised characters, with a larger absolute reduction for photorealistic rendering."
Paraphrase used:	Lighting structure of toon shading (lighter mid-tones) inherently aligns with low-eeriness configurations; extends the mechanism for why NPR reduces eeriness beyond shading style alone.
Original source text:	""Increasing key light brightness significantly increased appeal ($F = 14.3$, $p < .001$) and decreased eeriness ($F = 9.7$, $p < .001$). Eeriness reduction was larger for CG rendering than for toon. Toon characters were already less eerie at baseline." (Article summary; DOI: 10.1145/3383195.)"
Notes:	Source: Wisessing et al. (2020), *ACM Transactions on Graphics*, 39(3), Article 30. DOI: 10.1145/3383195. Verified from article_summaries.tex. NEEDS PHYSICAL PDF VERIFICATION.

Validated — original PDF read; paraphrase confirmed accurate.

#96 mcdonnell2009 — McDonnell et al. (2009) *§2.1 Uncanny Valley — emotion perception robust to avatar body style*

Cited sentence (claim in thesis):	"A companion study by the same group further demonstrated that the perception of emotional actions portrayed by animated characters is largely independent of avatar body style, remaining consistent across photorealistic, cartoon, and stylised skeletal representations."
Paraphrase used:	Supports the claim that NPR abstraction preserves communicative legibility of avatar behaviour; emotion reading does not require photorealism.
Original source text:	""The perception of emotional actions is highly robust and to the most part independent of the character's body, so long as form is present." (Physical PDF confirmed: McDonnell09_HappyZombies.pdf — ACM Reference: McDonnell et al., 2009, DOI: 10.1145/1577755.1577757.)"
Notes:	Source: McDonnell et al. (2009), *ACM Trans. Appl. Percept.*, 6(3), Article 14. DOI: 10.1145/1577755.1577757. Physical PDF CONFIRMED in Articles folder. Verified quote from abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#97 kyprianidis2011 — Kyprianidis (2011) *§2.3 NPR — multiscale extension of anisotropic Kuwahara filter*

Cited sentence (claim in thesis):	"A multiscale variant of this filter, also by Kyprianidis, propagates orientation estimates and filter outputs through a resolution pyramid, removing artefacts in low-contrast regions while maintaining real-time performance."
Paraphrase used:	Contextualises kyprianidis2009 anisotropic Kuwahara with its multiscale follow-up; both cited for the V6/V7/V8 Kuwahara technique family.
Original source text:	""A multi-scale computation scheme is proposed that simultaneously propagates local orientation estimates and filtering results up a low-pass filtered pyramid. This allows for a strong abstraction effect and avoids artifacts in large low-contrast regions." (Physical PDF confirmed: anisotropic_multicale_209.pdf — NPAR 2011, p. 55.)"
Notes:	Source: Kyprianidis (2011), NPAR '11, pp. 55–64. ACM ISBN 978-1-4503-0907-3. Physical PDF CONFIRMED in Articles/NPR folder. Quote verified from abstract on p. 1 of the PDF.

Validated — original PDF read; paraphrase confirmed accurate.